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TRACKER MODULE AND COMMUNICATION DEVICE

Abstract

An ET module is provided that includes a module laminate; first and second external output terminals disposed on the module laminate (90); and a voltage supply circuit disposed on the module laminate and configured to supply a power supply voltage $V_{sub.ETA}$ based on an envelope signal to a PA module via the first external output terminal, and to simultaneously supply a power supply voltage $V_{sub.ETB}$ based on an envelope signal to a PA module via the second external output terminal. The module laminate has first and second sides adjacent to each other, the first side and the second side forming at least a part of an outer shape of the module laminate in a plan view thereof. The first external output terminal is disposed adjacent to the first side, and the second external output terminal is disposed adjacent to the second side but not the first side.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of International Application No. PCT/JP2023/044470, filed Dec. 12, 2023, which claims priority to Japanese Patent Application No. 2022-206005, filed Dec. 22, 2022, the contents of each of which are hereby incorporated by reference in their entirety.

TECHNICAL FIELD

[0002] The exemplary aspects of the present disclosure relate to a tracker module and a communication device.

BACKGROUND

[0003] In recent years, a power added efficiency (hereinafter referred to as “efficiency”) has been improved by application of envelope tracking (ET: Envelope Tracking) to a power amplifier circuit. In ET systems, for example, a digital ET technology for supplying a plurality of discrete voltages has been disclosed.

[0004] For example, U.S. Pat. No. 9,755,672 discloses a supply modulator (e.g., a tracker circuit) that supplies a power supply voltage to a power amplifier circuit based on an envelope signal. The above-mentioned supply modulator includes a magnetic converter circuit (Magnetic Regulation Stage: Pre-regulator circuit) that converts a voltage; a switched-capacitor circuit (Switched-Capacitor Voltage Balance Stage) that generates, from the voltage, a plurality of voltages having different voltage levels; and an output switching circuit (Output Switching Stage) that selects and outputs at least one of the plurality of voltages. The magnetic converter circuit includes a switch and a power inductor, the switched-capacitor circuit includes a switch and a capacitor, and the output switching circuit includes a switch.

[0005] However, if a tracker circuit is modularized in an ET system, efficiency of the tracker circuit may deteriorate depending on an arrangement configuration of a voltage output path that outputs a voltage from the tracker circuit to a power amplifier circuit and a voltage input path that inputs the above voltage from the voltage output path to the tracker circuit.

SUMMARY OF THE INVENTION

[0006] In view of the foregoing, the exemplary aspects of the present disclosure provide a tracker module and a communication device that suppresses efficiency deterioration.

[0007] In an exemplary aspect, a tracker module is provided that includes a module laminate; a first external output terminal and a second external output terminal disposed on the module laminate; and a voltage supply circuit disposed on the module laminate and configured to supply a first voltage based on an envelope signal to a first power amplifier via the first external output terminal, and to simultaneously supply a second voltage based on an envelope signal to a second power amplifier via the second external output terminal. The module laminate has a first side and a second side adjacent to the first side. The first side and the second side form at least a part of an outer shape of the module laminate in a plan view of the module laminate. The first external output terminal is disposed adjacent to the first side, and the second external output terminal is disposed adjacent to the second side but not the first side.

[0008] In another exemplary aspect, a tracker module is provided that includes a module laminate;

a first external output terminal and a second external output terminal disposed on the module laminate; and a voltage supply circuit including at least one integrated circuit disposed on the module laminate. The at least one integrated circuit includes at least one switch included in a switched-capacitor circuit, at least one switch included in a first output switching circuit, and at least one switch included in a second output switching circuit. The switched-capacitor circuit is configured to generate a plurality of discrete voltages based on an input voltage and to output the generated plurality of discrete voltages to the first output switching circuit and the second output switching circuit. The first output switching circuit is configured to selectively output at least one discrete voltage of the generated plurality of discrete voltages to a first power amplifier via the first external output terminal, and the second output switching circuit is configured to selectively output at least one discrete voltage of the generated plurality of discrete voltages to a second power amplifier via the second external output terminal. The module laminate has a first side and a second side adjacent to the first side. The first side and the second side form at least a part of an outer shape of the module laminate in plan view of the module laminate. The first external output terminal is disposed adjacent to the first side, and the second external output terminal is disposed adjacent to the second side but not the first side.

[0009] According to the exemplary aspects of the present disclosure, a tracker module and a communication device are provided in which efficiency deterioration is suppressed.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0010] FIG. 1A is a graph illustrating an example of a transition in a power supply voltage in an average power tracking (APT: Average Power Tracking) mode.

[0011] FIG. 1B is a graph illustrating an example of a transition in a power supply voltage in an analog ET mode.

[0012] FIG. 1C is a graph illustrating an example of a transition in a power supply voltage in a digital ET mode.

[0013] FIG. 2 is a circuit configuration diagram of a tracker module and a communication device according to an exemplary embodiment.

[0014] FIG. 3 is a circuit configuration diagram of a tracker module and a communication device according to Example 1.

[0015] FIG. 4 is a diagram of circuit configurations of a pre-regulator circuit, a switched-capacitor circuit, output switching circuits, and filter circuits according to Example 1.

[0016] FIG. 5 is a circuit configuration diagram of a tracker module and a communication device according to Example 2.

[0017] FIG. 6A is a plan view of the tracker module according to Example 1.

[0018] FIG. 6B is a plan view of the tracker module according to Example 1.

[0019] FIG. 6C is a cross-sectional view of the tracker module according to Example 1.

[0020] FIG. 7 is a plan view of the communication device according to Example 1.

[0021] FIG. 8 is a plan view of the communication device according to Example 2.

[0022] FIG. 9 is a plan view of a communication device according to a modification example.

DETAILED DESCRIPTION OF EMBODIMENTS

[0023] In the following, embodiments of the present disclosure are described in detail with reference to the drawings. All embodiments to be described below illustrate comprehensive and specific examples. Numeric values, shapes, materials, components, and an arrangement and a connection form of the components, which are described in the following embodiments, are merely examples and are not intended to limit the exemplary aspects of the present disclosure.

[0024] It is noted that each figure is a schematic diagram that is appropriately emphasized, omitted,

or subjected to ratio adjustment in order to illustrate the exemplary aspects of the present disclosure. Each figure is not necessarily a strict illustration and may differ from an actual shape, a positional relationship, or a ratio. In each figure, an identical numeral is given to a substantially identical configuration, and an overlapping description may be omitted or simplified.

[0025] In each figure below, an x-axis and a y-axis are axes orthogonal to each other on a plane parallel to a main surface of a board. Specifically, if a board has a rectangular shape in plan view, the x-axis is parallel to a first side of the board, and the y-axis is parallel to a second side that is orthogonal to the first side of the board. In addition, a z-axis is an axis perpendicular to the main surface of the board, a positive direction of which indicates an upward direction and a negative direction of which indicates a downward direction.

[0026] In a circuit configuration of the present disclosure, the phrase “to be connected” includes a case where a connection terminal and/or a wiring conductor is directly connected, and a case in which a connection terminal and/or a wiring conductor is electrically connected via another circuit element. Moreover, the phrase “to be connected between A and B” can refer to a connection with both A and B between A and B.

[0027] In a component arrangement of the present disclosure, the phrase “a component disposed in or on a board” includes a component being disposed on a main surface of a board and a component being disposed in a board. Moreover, the phrase “a component being disposed on a main surface of a board” includes not only a component being disposed in contact with a main surface of a board, but also a component being disposed above the main surface without being in contact therewith (for example, a component being laminated on another component that is disposed in contact with a main surface). In addition, the phrase “a component being disposed on a main surface of a board” may include a component being disposed in a recessed portion formed on a main surface. The phrase “a component being disposed in a board” includes not only a component being encapsulated within a board but also an entire component being disposed between both main surfaces of a board, but a portion of the component not being covered with the board, and only a portion of the component being disposed in the board.

[0028] In the component arrangement of the present disclosure, the phrase “in plan view of a board” can mean viewing, from a z-axis positive side, an object or component orthographically projected on an xy-plane. The phrase “A overlapping with B in plan view” can mean at least a portion of a region of A orthographically projected on the xy-plane overlaps with at least a portion of a region of B orthographically projected on the xy-plane. In addition, the phrase “A being disposed between B and C” can mean that at least one of a plurality of line segments connecting an arbitrary point in B and an arbitrary point in C passes through A.

[0029] In addition, in the component arrangement of the present disclosure, the phrase “A being disposed adjacent to B” represents that A and B are disposed in proximity, and specifically can mean that no other circuit component is present in a space in which A faces B. In other words, “A being disposed adjacent to B” can mean that none of a plurality of line segments reaching B, from an arbitrary point on a surface of A facing B, along a normal direction of the surface, passes through any circuit component other than A and B. Here, a circuit component can refer to a component including an active element and/or a passive element. More specifically, a circuit component includes an active component including a transistor or a diode, or the like, and a passive component including an inductor, a transformer, a capacitor, or a resistor, or the like, and does not include an electromechanical component including a terminal, a connector, or a wiring line, or the like.

[0030] In the exemplary aspects of the present disclosure, a “terminal” can refer to a point where a conductor within an element terminates. Moreover, it is noted that when impedance of a conductor between elements is sufficiently low, a terminal is construed not only as a single point but also as an arbitrary point on the conductor between the elements or the entire conductor.

[0031] In the component arrangement of the present disclosure, the phrase “component A being

disposed in series on path B” can mean that both a signal input terminal and a signal output terminal of the component A are connected to a wiring line, an electrode, or a terminal that form the path B.

[0032] In addition and for purposes of this disclosure, terms that indicate a relationship between elements, such as “parallel” or “perpendicular”, and terms that indicate a shape of an element, such as “rectangular”, as well as a numerical range do not only indicate a strict meaning, but also may include a substantially equivalent range, for example, an error such as a few percent.

[0033] First, as a technology for amplifying radio frequency signals highly efficiently, a tracking mode is described that supplies a power amplifier with a power supply voltage to be dynamically adjusted over time based on the radio frequency signal. The tracking mode is a mode that dynamically adjusts a power supply voltage to be applied to the power amplifier. Although there are several types of tracking modes, here, an average power tracking (APT: Average Power Tracking) mode and an ET (ET: Envelope Tracking mode (including an analog ET mode and a digital ET mode) are described with reference to FIGS. 1A to 1C. In FIGS. 1A to 1C, a horizontal axis represents time and a vertical axis represents a voltage. In addition, a thick solid line represents a power supply voltage and a thin solid line (waveform) represents a modulated signal.

[0034] FIG. 1A is a graph illustrating an example of a transition in a power supply voltage in the APT mode. In the APT mode, a power supply voltage is fluctuated to a plurality of discrete voltage levels in units of one frame based on average power. As a result, a power supply voltage signal forms a rectangular wave.

[0035] According to an exemplary aspect, a frame can refer to a unit that forms a radio frequency signal (e.g., a modulated signal). For example, in 5G NR (5.sup.th Generation New Radio) and LTE (Long Term Evolution), a frame includes 10 subframes, each subframe including a plurality of slots, each slot being composed of a plurality of symbols. A subframe has a length of 1 ms, and a frame has a length of 10 ms.

[0036] It is noted that a mode of fluctuating a voltage level in units of one frame or larger based on the average power is referred to as the APT mode, and is distinguished from a mode that fluctuates the voltage level in units smaller than one frame (for example, a subframe, a slot, or a symbol). For example, a mode that fluctuates the voltage level in symbol units is referred to as a symbol power tracking (SPT: Symbol Power Tracking) mode and is distinguished from the APT mode.

[0037] FIG. 1B is a graph illustrating an example of a transition in a power supply voltage in an analog ET mode. In the analog ET mode, an envelope of the modulated signal is tracked by continuously fluctuating the power supply voltage based on an envelope signal.

[0038] An envelope signal is a signal representing an envelope of a modulated signal. An envelope value is expressed by, for example, a root square of $(I_{\text{sup}}^2 + Q_{\text{sup}}^2)$. Here, (I, Q) represents a constellation point. A constellation point refers to a point that represents a signal modulated by digital modulation on a constellation diagram. (I, Q) is determined by, for example, BBIC (Baseband Integrated Circuit) based on, for example, transmission information.

[0039] FIG. 1C is a graph illustrating an example of a transition in a power supply voltage in a digital ET mode. In the digital ET mode, an envelope of a modulated signal is tracked by fluctuating the power supply voltage to a plurality of discrete voltage levels in one frame based on an envelope signal. As a result, the power supply voltage signal forms a rectangular wave.

Exemplary Embodiments

[1. Circuit Configuration of ET Module 1 and Communication Device 5]

[0040] An ET module 1 and a communication device 5 according to the present embodiment are described with reference to FIG. 2.

[0041] FIG. 2 is a circuit configuration diagram of the ET module 1 and the communication device 5 according to the present embodiment. The communication device 5 according to the present embodiment corresponds to a user terminal (UE User Equipment) in a cellular network, and is typically a mobile phone, a smart phone, a tablet computer, a wearable device, or the like. It is

noted that the communication device **5** may be an IoT (Internet of Things) sensor device, a medical/healthcare device, an automobile, an unmanned aerial vehicle (UAV: Unmanned Aerial Vehicle) (a so-called drone), and an automated guided vehicle (AGV: Automated Guided Vehicle). [0042] First, a circuit configuration of the communication device **5** is described. As illustrated in FIG. **2**, the communication device **5** according to the present embodiment includes the ET module **1**, PA modules **2A** and **2B**, antennae **3A** and **3B**, and an RFIC (Radio Frequency Integrated Circuit) **4**.

[0043] The antenna **3A** is connected to the PA module **2A** and transmits a radio frequency signal output from the PA module **2A**. The antenna **3B** is connected to the PA module **2B** and transmits a radio frequency signal output from the PA module **2B**.

[0044] The RFIC **4** is an example of a signal processing circuit that processes radio frequency signals. The RFIC **4** has a control section that controls the PA modules **2A** and **2B**. Specifically, the RFIC **4** performs signal processing on a transmission signal input from the BBIC (not illustrated) by up-conversion, or the like, and outputs a radio frequency transmission signal generated by the signal processing to the PA modules **2A** and **2B**. In addition, the RFIC **4** outputs a digital control signal that controls the ET module **1** to the ET module **1**. It is noted that some or all of capabilities of the RFIC **4** as the control section may be implemented externally to the RFIC **4**, and may be implemented, for example, in the BBIC, the PA modules **2A** and **2B**, and the ET module **1**.

[0045] The ET module **1** supplies a power supply voltage $V_{sub.ETA}$ to the PA module **2A** and supplies a power supply voltage $V_{sub.ETB}$ to the PA module **2B**. The power supply voltage $V_{sub.ETA}$ and the power supply voltage $V_{sub.ETB}$ are examples of voltages based on envelope signals. The ET module **1** can supply the power supply voltage $V_{sub.ETA}$ to the PA module **2A** and supply the power supply voltage $V_{sub.ETB}$ to the PA module **2B** in at least any of the APT mode, the analog ET mode, and the digital ET mode illustrated in FIGS. **1A** to **1C**.

[0046] As illustrated in FIG. **2**, the ET module **1** includes a voltage supply circuit **10**, a control circuit **70**, external output terminals **101** and **104**, external input terminals **102** and **105**, and control signal terminal **103**.

[0047] The voltage supply circuit **10** is configured to supply the power supply voltage $V_{sub.ETA}$ based on the envelope signal to the PA module **2A** by way of the external output terminal **101**, and to supply the power supply voltage $V_{sub.ETB}$ based on the envelope signal to the PA module **2B** by way of an external output terminal **104**. Specifically, the voltage supply circuit **10** receives a control signal output from the control circuit **70** to generate the power supply voltage $V_{sub.ETA}$, and outputs the power supply voltage $V_{sub.ETA}$ to the external output terminal **101**. In addition, the voltage supply circuit **10** receives a control signal output from the control circuit **70** to generate the power supply voltage $V_{sub.ETB}$, and outputs the power supply voltage $V_{sub.ETB}$ to the external output terminal **104**. The voltage supply circuit **10** is, for example, a DC-DC converter that steps up or down a pressure of magnetic energy generated by an inductor of the DC-DC converter and outputs the magnetic energy, as the power supply voltage $V_{sub.ETA}$ and the power supply voltage $V_{sub.ETB}$. It is noted that the inductor that generates the magnetic energy may be located externally to the ET module **1** in an exemplary aspect.

[0048] The external output terminal **101** is an example of a first external output terminal that can be connected externally and is connected to the voltage supply circuit **10** and the PA module **2A**. The power supply voltage $V_{sub.ETA}$, which is generated by the voltage supply circuit **10**, is applied to the external output terminal **101**.

[0049] The external output terminal **104** is an example of a second external output terminal that can be connected externally and is connected to the voltage supply circuit **10** and the PA module **2B**. The power supply voltage $V_{sub.ETB}$, which is generated by the voltage supply circuit **10**, is applied to the external output terminal **104**.

[0050] The external input terminal **102** is an example of a first external input terminal that can be connected externally and is connected to a first path connecting the external output terminal **101**

and the PA module **2A**. In operation, when the analog ET mode is applied to the ET module **1**, the external input terminal **102** is configured to function as a terminal for feeding back the power supply voltage $V_{sub.ETA}$ on the first path to the ET module **1**, and is connected to the voltage supply circuit **10** or the control circuit **70** within the ET module **1**. In addition, when the APT mode is applied to the ET module **1**, the external input terminal **102** is configured to function as a terminal for stabilizing the power supply voltage $V_{sub.ETA}$ on the first path, and is connected to a capacitor, for example.

[0051] The external input terminal **105** is an example of the second external input terminal that can be connected externally and is connected to a second path connecting the external output terminal **104** and the PA module **2B**. In operation, when the analog ET mode is applied to the ET module **1**, the external input terminal **105** is configured to function as a terminal for feeding back the power supply voltage $V_{sub.ETB}$ on the second path to the ET module **1**, and is connected to the voltage supply circuit **10** or the control circuit **70** within the ET module **1**. In addition, when the APT mode is applied to the ET module **1**, the external input terminal **105** is configured to function as a terminal for stabilizing the power supply voltage $V_{sub.ETB}$ on the second path, and is connected to a capacitor, for example.

[0052] The control circuit **70** is an example of a control section of the ET module **1** and is connected to the control signal terminal **103** and the voltage supply circuit **10**. The control circuit **70** controls the voltage supply circuit **10** to output the power supply voltage $V_{sub.ETA}$ and the power supply voltage $V_{sub.ETB}$ from the voltage supply circuit **10**, based on an envelope signal input from the RFIC **4** via the control signal terminal **103**, target power supply voltage information of the PA modules **2A** and **2B**, and a feedback power supply voltage input via the external input terminals **102** and **105**.

[0053] The PA module **2A** receives supply of the power supply voltage $V_{sub.ETA}$ from the ET module **1**, amplifies a radio frequency transmission signal input from the RFIC **4**, and outputs the amplified radio frequency transmission signal to the antenna **3A**. The PA module **2A** includes power amplifiers **41**, **42**, and **43**, filters **51**, **52**, and **53**, and a switch **61**.

[0054] Each of the power amplifiers **41** to **43** is an example of a first power amplifier and is connected between the ET module **1** and the antenna **3A**. Specifically, a supply terminal of the power amplifier **41** is connected to the ET module **1**, a radio frequency input terminal is connected to the RFIC **4**, and a radio frequency output terminal is connected to the antenna **3A** via the filter **51** and the switch **61**. A supply terminal of the power amplifier **42** is connected to the ET module **1**, a radio frequency input terminal is connected to the RFIC **4**, and a radio frequency output terminal is connected to the antenna **3A** via the filter **52** and the switch **61**. A supply terminal of the power amplifier **43** is connected to the ET module **1**, a radio frequency input terminal is connected to the RFIC **4**, and a radio frequency output terminal is connected to the antenna **3A** via the filter **53** and the switch **61**. The power amplifier **41** can amplify a radio frequency signal of band A, the power amplifier **42** can amplify a radio frequency signal of band B, and the power amplifier **43** can amplify a radio frequency signal of band C.

[0055] The filter **51** includes the band A as a pass band, for example. The filter **52** includes the band B as a pass band, for example. The filter **53** includes the band C as a pass band, for example.

[0056] The switch **61** has a common terminal and three selection terminals, and switches connection between the common terminal and any one of the three selection terminals. The common terminal of the switch **61** is connected to the antenna **3A**, a first selection terminal is connected to the filter **51**, a second selection terminal is connected to the filter **52**, and a third selection terminal is connected to the filter **53**.

[0057] According to a configuration of the PA module **2A** described above, the PA module **2A** can amplify a radio frequency signal of any one of the band A, the band B, and the band C, and output the amplified radio frequency signal to the antenna **3A**.

[0058] A node on the first path connecting each of the supply terminals of the power amplifiers **41**

to **43** and the external output terminal **101** is connected to the external input terminal **102**.

[0059] The PA module **2B** receives supply of the power supply voltage $V_{sub.ETB}$ from the ET module **1**, amplifies a radio frequency transmission signal input from the RFIC **4**, and outputs the amplified radio frequency transmission signal to the antenna **3B**. The PA module **2B** includes power amplifiers **44**, **45** and **46**, filters **54**, **55**, and **56**, and a switch **62**.

[0060] Each of the power amplifiers **44** to **46** is an example of a second power amplifier and is connected between the ET module **1** and the antenna **3B**. Specifically, a supply terminal of the power amplifier **44** is connected to the ET module **1**, a radio frequency input terminal is connected to the RFIC **4**, and a radio frequency output terminal is connected to the antenna **3B** via the filter **54** and the switch **62**. A supply terminal of the power amplifier **45** is connected to the ET module **1**, a radio frequency input terminal is connected to the RFIC **4**, and a radio frequency output terminal is connected to the antenna **3B** via the filter **55** and the switch **62**. A supply terminal of the power amplifier **46** is connected to the ET module **1**, a radio frequency input terminal is connected to the RFIC **4**, and a radio frequency output terminal is connected to the antenna **3B** via the filter **56** and the switch **62**. The power amplifier **44** can amplify a radio frequency signal of band D, the power amplifier **45** can amplify a radio frequency signal of band E, and the power amplifier **46** can amplify a radio frequency signal of band F.

[0061] The filter **54** includes the band D as a pass band, for example. The filter **55** includes the band E as a pass band, for example. The filter **56** includes the band F as a pass band, for example.

[0062] The switch **62** has a common terminal and three selection terminals, and switches connection between the common terminal and any one of the three selection terminals. The common terminal of the switch **62** is connected to the antenna **3B**, a first selection terminal is connected to the filter **54**, a second selection terminal is connected to the filter **55**, and a third selection terminal is connected to the filter **56**.

[0063] According to a configuration of the PA module **2B** described above, the PA module **2B** can amplify a radio frequency signal of any one of the band D, the band E, and the band F, and output the amplified radio frequency signal to the antenna **3B**.

[0064] A node on the second path connecting each of the supply terminals of the power amplifiers **44** to **46** and the external output terminal **104** is connected to the external input terminal **105**.

[0065] It is noted that each of the band A to the band F is a frequency band for a communication system that is built using a radio access technology (RAT: Radio Access Technology), and is predefined by a standardizing body (for example, 3GPP® (3rd Generation Partnership Project) and IEEE (Institute of Electrical and Electronics Engineers), or the like). Examples of communication systems include a 5G NR (5.sup.th Generation New Radio) system, an LTE (Long Term Evolution) system, and a WLAN (Wireless Local Area Network) system, or the like.

[0066] According to the above-described configuration, the communication device **5** can simultaneously transmit a radio frequency signal of any of the band A to the band C and a radio frequency signal of any of the band D to the band F. In order to cope with this, the ET module **1** is configured to supply the power supply voltage $V_{sub.ETA}$ (e.g., a first voltage) based on the envelope signal to the PA module **2A** via the external output terminal **101**, and simultaneously supply the power supply voltage $V_{sub.ETB}$ (e.g., a second voltage) based on the envelope signal to the PA module **2B** via the external output terminal **104**.

[2. Circuit Configuration of ET Module **1A** and Communication Device **5A** According to Example 1]

[0067] FIG. **3** is a circuit configuration diagram of an ET module **1A** and the communication device **5A** according to Example 1. The ET module **1A** and the communication device **5A** according to this Example are examples of the ET module **1** and the communication device **5** according to the present embodiment and illustrate a specific circuit configuration of the ET module **1**.

[0068] The communication device **5A** according to this Example includes the ET module **1A**, the

PA modules **2A** and **2B**, capacitors **31A** and **31B**, antennae **3A** and **3B**, and the RFIC **4**. As compared to the communication device **5** according to the embodiment, the communication device **5A** according to this Example differs in that a configuration of the ET module **1A** as well as the capacitors **31A** and **31B** are added. Therefore, in the following, the communication device **5A** according to this Example is described with a focus on configurations of the ET module **1A** and the capacitors **31A** and **31B**.

[0069] The ET module **1A** is an example of a tracker module, and can supply the power supply voltage $V_{\text{sub.ETA}}$, which is one of a plurality of discrete voltages, to the PA module **2A**, and supply the power supply voltage $V_{\text{sub.ETB}}$, which is one of the plurality of discrete voltages, to the PA module **2B**. The digital ET mode and the APT mode can be used as a tracking mode, but the tracking mode is not limited thereto. As illustrated in FIG. **3**, the ET module **1A** includes a voltage supply circuit **10A**, the control circuit **70**, switches **21A** and **21B**, the external output terminals **101** and **104**, the external input terminals **102** and **105**, and the control signal terminal **103**. As compared to the ET module **1** according to the embodiment, the ET module **1A** according to this Example differs in that a configuration of the voltage supply circuit **10A** and the switches **21A** and **21B** are added. In the following, the ET module **1A** according to this Example is described, omitting the same configurations as those of the ET module **1** according to the embodiment and focusing on different configurations.

[0070] The voltage supply circuit **10A** includes a pre-regulator circuit **11**, a switched-capacitor circuit **12**, output switching circuits **13A** and **13B**, and filter circuits **14A** and **14B**.

[0071] The pre-regulator circuit **11** is an example of a converter circuit and includes a power inductor and a switch. A power inductor is an inductor used to step up and/or down a direct current (DC: Direct Current) voltage. The power inductor is connected in series to a direct current path. It is noted that the power inductor may be connected (e.g., disposed in parallel) between the direct current path and a ground according to an exemplary aspect. The pre-regulator circuit **11** can use the power inductor to convert an input voltage into a third voltage. In some cases, such a pre-regulator circuit **11** may be referred to as a magnetic regulator or a DC-DC converter.

[0072] The switched-capacitor circuit **12** includes a plurality of capacitors and a plurality of switches, and can be configured to generate a plurality of discrete voltages from input voltages from the pre-regulator circuit **11**. The switched-capacitor circuit **12** may be referred to as a switched-capacitor voltage balancer (Switched-Capacitor Voltage Balancer).

[0073] The output switching circuit **13A** is an example of a first output switching circuit and is configured to selectively output at least one of the plurality of discrete voltages generated by the switched-capacitor circuit **12** to the PA module **2A** via the external output terminal **101**. The output switching circuit **13A** is controlled based on a digital control signal.

[0074] The output switching circuit **13B** is an example of a second output switching circuit and is configured to selectively output at least one of the plurality of discrete voltages generated by the switched-capacitor circuit **12** to the PA module **2B** via the external output terminal **104**. The output switching circuit **13B** is controlled based on a digital control signal.

[0075] The filter circuit **14A** can attenuate noise from the plurality of discrete voltages supplied to the PA module **2A**. The filter circuit **14B** can attenuate noise from the plurality of discrete voltages supplied to the PA module **2B**. The filter circuits **14A** and **14B** may be referred to as pulse shaping filters or transition shaping filters.

[0076] It is noted that the voltage supply circuit **10A** may omit the filter circuits **14A** and **14B** in an exemplary aspect. In addition, any combination of the pre-regulator circuit **11**, the switched-capacitor circuit **12**, the output switching circuits **13A** and **13B**, and the filter circuits **14A** and **14B** may be integrated into a single circuit. Alternatively, the voltage supply circuit **10A** may include a plurality of voltage supply circuits, instead of the pre-regulator circuit **11** and the switched-capacitor circuit **12**. In this case, the output switching circuits **13A** and **13B** may be each configured to select at least one of the plurality of voltage supply circuits.

[0077] The switch **21A** is an example of a first switch circuit and is connected between the external input terminal **102** and the ground to switch between connection and disconnection between the external input terminal **102** and the ground. The switch **21B** is connected between the external input terminal **105** and the ground, and switches between connection and disconnection between the external input terminal **105** and the ground.

[0078] The capacitor **31A** is disposed in series between the external input terminal **102** and a node on the first path. The capacitor **31B** is disposed in series between the external input terminal **105** and a node on the second path.

[0079] In operation, when the digital ET mode is applied to the ET module **1A**, for example, the switches **21A** and **21B** are in a non-conductive state, so that a radio frequency signal of a wide channel band width (wide modulation band width) can be amplified highly efficiently in the PA modules **2A** and **2B**. In addition, when the APT mode is applied to the ET module **1A**, for example, the switches **21A** and **21B** are in a conductive state, so that the capacitors **31A** and **31B** can be configured to function as bypass condensers, and voltage levels of the power supply voltage $V_{sub.ETA}$ and the power supply voltage $V_{sub.ETB}$ can be stabilized.

[0080] Next, circuit configuration of the pre-regulator circuit **11**, the switched-capacitor circuit **12**, the output switching circuits **13A** and **13B**, and the filter circuits **14A** and **14B**, which are included in the voltage supply circuit **10A**, are described with reference to FIG. **4**.

[0081] FIG. **4** is a diagram of circuit configurations of the pre-regulator circuit **11**, the switched-capacitor circuit **12**, the output switching circuits **13A** and **13B**, and filter circuits **14A** and **14B** according to Example 1.

[0082] It is noted that FIG. **4** is an exemplary circuit configuration, and that the pre-regulator circuit **11**, the switched-capacitor circuit **12**, the output switching circuits **13A** and **13B**, and the filter circuits **14A** and **14B** may be implemented using any of a wide variety of circuit implementations or circuit technologies. Therefore, a description of each of the circuits to be provided below should not be interpreted in a limited manner.

[2.1 Circuit Configuration of Switched-Capacitor Circuit **12**]

[0083] As illustrated in FIG. **4**, the switched-capacitor circuit **12** includes capacitors **C11** to **C16**, capacitors **C10**, **C20**, **C30**, and **C40**, and switches **S11** to **S14**, **S21** to **S24**, **S31** to **S34**, and **S41** to **S44**. Energy and electric charges are input from the pre-regulator circuit **11** to the switched-capacitor circuit **12** at nodes **N1** to **N4** and drawn from the switched-capacitor circuit **12** to the output switching circuits **13A** and **13B** at the nodes **N1** to **N4**.

[0084] Each of the capacitors **C11** to **C16** is configured to function as a flying capacitor (which may be referred to as a transfer capacitor) in an exemplary aspect. That is, each of the capacitors **C11** to **C16** are used to step up or down input voltages supplied from the pre-regulator circuit **11**. More specifically, the capacitors **C11** to **C16** transfer electric charges between the capacitors **C11** to **C16** and the nodes **N1** to **N4** so that voltages **V1** to **V4** (voltages with respect to a ground potential), which satisfy $V1:V2:V3:V4=1:2:3:4$ at the four nodes **N1** to **N4**, are maintained. These voltages **V1** to **V4** correspond to a plurality of discrete voltages.

[0085] The capacitor **C11** has two electrodes. One of the two electrodes of the capacitor **C11** is connected to one end of the switch **S11** and one end of the switch **S12**. The other of the two electrodes of the capacitor **C11** is connected to one end of the switch **S21** and one end of the switch **S22**.

[0086] The capacitor **C12** has two electrodes. One of the two electrodes of the capacitor **C12** is connected to the one end of the switch **S21** and the one end of the switch **S22**. The other of the two electrodes of the capacitor **C12** is connected to one end of the switch **S31** and one end of the switch **S32**.

[0087] The capacitor **C13** has two electrodes. One of the two electrodes of the capacitor **C13** is connected to the one end of the switch **S31** and one end of the switch **S32**. The other of the two electrodes of the capacitor **C13** is connected to one end of the switch **S41** and one end of the switch

S42.

[0088] The capacitor **C14** has two electrodes. One of the two electrodes of the capacitor **C14** is connected to one end of the switch **S13** and one end of the switch **S14**. The other of the two electrodes of the capacitor **C14** is connected to one end of the switch **S23** and one end of the switch **S24**.

[0089] The capacitor **C15** has two electrodes. One of the two electrodes of the capacitor **C15** is connected to the one end of the switch **S23** and the one end of the switch **S24**. The other of the two electrodes of the capacitor **C15** is connected to one end of the switch **S33** and one end of the switch **S34**.

[0090] The capacitor **C16** has two electrodes. One of the two electrodes of the capacitor **C16** is connected to the one end of the switch **S33** and the one end of the switch **S34**. The other of the two electrodes of the capacitor **C16** is connected to one end of the switch **S43** and one end of the switch **S44**.

[0091] Each of a set of the capacitors **C11** and **C14**, a set of the capacitors **C12** and **C15**, and a set of the capacitors **C13** and **C16** can perform charging and discharging in a complementary manner, with a first phase and a second phase being repeated.

[0092] Specifically, in the first phase, the switches **S12**, **S13**, **S22**, **S23**, **S32**, **S33**, **S42**, and **S43** are turned ON. As a result, for example, the one of the two electrodes of the capacitor **C12** is connected to the node **N3**, the other of the two electrodes of the capacitor **C12** and the one of the two electrodes of the capacitor **C15** are connected to the node **N2**, and the other of the two electrodes of the capacitor **C15** is connected to the node **N1**.

[0093] In contrast, in the second phase, the switches **S11**, **S14**, **S21**, **S24**, **S31**, **S34**, **S41**, and **S44** are turned ON. As a result, for example, the one of the two electrodes of the capacitor **C15** is connected to the node **N3**, the other of the two electrodes of the capacitor **C15** and the one of the two electrodes of the capacitor **C12** are connected to the node **N2**, and the other of the two electrodes of the capacitor **C12** is connected to the node **N1**.

[0094] Repetition of such first and second phases enables the other of the capacitor **C12** and the other of the capacitor **C15** to discharge to the capacitor **C30**, for example, while the one of the capacitor **C12** and one of the capacitor **C15** are being charged from the node **N2**. That is, the capacitors **C12** and **C15** can perform charging and discharging in a complementary manner.

[0095] Similarly to the set of the capacitors **C12** and **C15**, the repetition of the first and second phases enables each of the set of the capacitors **C11** and **C14** and the set of the capacitors **C13** and **C16** to perform charging and discharging in a complementary manner.

[0096] Each of the capacitors **C10**, **C20**, **C30**, and **C40** is configured to function as a smoothing capacitor in an exemplary aspect. That is, each of the capacitors **C10**, **C20**, **C30**, and **C40** is used to hold and smoothen the voltages **V1** to **V4** at the nodes **N1** to **N4**.

[0097] The capacitor **C10** is connected between the node **N1** and the ground. Specifically, one of two electrodes of the capacitor **C10** is connected to the node **N1**. Meanwhile, the other of the two electrodes of the capacitor **C10** is connected to the ground.

[0098] The capacitor **C20** is connected between the nodes **N2** and **N1**. Specifically, one of two electrodes of the capacitor **C20** is connected to the node **N2**. Meanwhile, the other of the two electrodes of the capacitor **C20** is connected to the node **N1**.

[0099] The capacitor **C30** is connected between the nodes **N3** and **N2**. Specifically, one of two electrodes of the capacitor **C30** is connected to the node **N3**. Meanwhile, the other of the two electrodes of the capacitor **C30** is connected to the node **N2**.

[0100] The capacitor **C40** is connected between the nodes **N4** and **N3**. Specifically, one of two electrodes of the capacitor **C40** is connected to the node **N4**. Meanwhile, the other of the two electrodes of the capacitor **C40** is connected to the node **N3**.

[0101] The switch **S11** is connected between the one of the two electrodes of the capacitor **C11** and the node **N3**. Specifically, the one end of the switch **S11** is connected to the one of the two

electrodes of the capacitor C11. Meanwhile, another end of the switch S11 is connected to the node N3.

[0102] The switch S12 is connected between the one of the two electrodes of the capacitor C11 and the node N4. Specifically, the one end of the switch S12 is connected to the one of the two electrodes of the capacitor C11. Meanwhile, another end of the switch S12 is connected to the node N4.

[0103] The switch S21 is connected between the one of the two electrodes of the capacitor C12 and the node N2. Specifically, the one end of the switch S21 is connected to the one of the two electrodes of the capacitor C12 and the other of the two electrodes of the capacitor C11.

Meanwhile, another end of the switch S21 is connected to the node N2.

[0104] The switch S22 is connected between the one of the two electrodes of the capacitor C12 and the node N3. Specifically, the one end of the switch S22 is connected to the one of the two electrodes of the capacitor C12 and the other of the two electrodes of the capacitor C11.

Meanwhile, another end of the switch S22 is connected to the node N3.

[0105] The switch S31 is connected between the other of the two electrodes of the capacitor C12 and the node N1. Specifically, the one end of the switch S31 is connected to the other of the two electrodes of the capacitor C12 and the one of the two electrodes of the capacitor C13. Meanwhile, another end of the switch S31 is connected to the node N1.

[0106] The switch S32 is connected between the other of the two electrodes of the capacitor C12 and the node N2. Specifically, the one end of the switch S32 is connected to the other of the two electrodes of the capacitor C12 and the one of the two electrodes of the capacitor C13. Meanwhile, another end of the switch S32 is connected to the node N2. That is, the other end of the switch S32 is connected to the other end of the switch S21.

[0107] The switch S41 is connected between the other of the two electrodes of the capacitor C13 and the ground. Specifically, the one end of the switch S41 is connected to the other of the two electrodes of the capacitor C13. Meanwhile, another end of the switch S41 is connected to the ground.

[0108] The switch S42 is connected between the other of the two electrodes of the capacitor C13 and the node N1. Specifically, the one end of the switch S42 is connected to the other of the two electrodes of the capacitor C13. Meanwhile, another end of the switch S42 is connected to the node N1. That is, the other end of the switch S42 is connected to the other end of the switch S31.

[0109] The switch S13 is connected between the one of the two electrodes of the capacitor C14 and the node N3. Specifically, the one end of the switch S13 is connected to the one of the two electrodes of the capacitor C14. Meanwhile, another end of the switch S13 is connected to the node N3. That is, the other end of the switch S13 is connected to the other end of the switch S11 and the other end of the switch S22.

[0110] The switch S14 is connected between the one of the two electrodes of the capacitor C14 and the node N4. Specifically, the one end of the switch S14 is connected to the one of the two electrodes of the capacitor C14. Meanwhile, another end of the switch S14 is connected to the node N4. That is, the other end of the switch S14 is connected to the other end of the switch S12.

[0111] The switch S23 is connected between the one of the two electrodes of the capacitor C15 and the node N2. Specifically, the one end of the switch S23 is connected to the one of the two electrodes of the capacitor C15 and the other of the two electrodes of the capacitor C14.

Meanwhile, another end of the switch S23 is connected to the node N2. That is, the other end of the switch S23 is connected to the other end of the switch S21 and the other end of the switch S32.

[0112] The switch S24 is connected between the one of the two electrodes of the capacitor C15 and the node N3. Specifically, the one end of the switch S24 is connected to the one of the two electrodes of the capacitor C15 and the other of the two electrodes of the capacitor C14.

Meanwhile, another end of the switch S24 is connected to the node N3. That is, the other end of the switch S24 is connected to the other end of the switch S11, the other end of the switch S22, and the

other end of the switch **S13**.

[0113] The switch **S33** is connected between the other of the two electrodes of the capacitor **C15** and the node **N1**. Specifically, the one end of the switch **S33** is connected to the other of the two electrodes of the capacitor **C15** and the one of the two electrodes of the capacitor **C16**. Meanwhile, another end of the switch **S33** is connected to the node **N1**. That is, the other end of the switch **S33** is connected to the other end of the switch **S31** and the other end of the switch **S42**.

[0114] The switch **S34** is connected between the other of the two electrodes of the capacitor **C15** and the node **N2**. Specifically, the one end of the switch **S34** is connected to the other of the two electrodes of the capacitor **C15** and the one of the two electrodes of the capacitor **C16**. Meanwhile, another end of the switch **S34** is connected to the node **N2**. That is, the other end of the switch **S34** is connected to the other end of the switch **S21**, the other end of the switch **S32**, and the other end of the switch **S23**.

[0115] The switch **S43** is connected between the other of the two electrodes of the capacitor **C16** and the ground. Specifically, the one end of the switch **S43** is connected to the other of the two electrodes of the capacitor **C16**. Meanwhile, another end of the switch **S43** is connected to the ground.

[0116] The switch **S44** is connected between the other of the two electrodes of the capacitor **C16** and the node **N1**. Specifically, the one end of the switch **S44** is connected to the other of the two electrodes of the capacitor **C16**. Meanwhile, another end of the switch **S44** is connected to the node **N1**. That is, the other end of the switch **S44** is connected to the other end of the switch **S31**, the other end of the switch **S42**, and the other end of the switch **S33**.

[0117] The switches **S11**, **S12**, **S13**, **S14**, **S21**, **S22**, **S23**, and **S24** are each at least one switch included in the switched-capacitor circuit **12**.

[0118] A first set of switches including the switches **S12**, **S13**, **S22**, **S23**, **S32**, **S33**, **S42**, and **S43** and a second set of switches including the switches **S11**, **S14**, **S21**, **S24**, **S31**, **S34**, **S41**, and **S44** are switched ON and OFF based on a control signal **S2** in a complementary manner. Specifically, in the first phase, the first set of switches is turned ON and the second set of switches is turned OFF. Conversely, in the second phase, the first set of switches is turned OFF and the second set of switches is turned ON.

[0119] For example, in one of the first phase and the second phase, charging of the capacitors **C10** to **C40** is performed from the capacitors **C11** to **C13**, and in the other of the first phase and the second phase, charging of the capacitors **C10** to **C40** is performed from the capacitors **C14** to **C16**. That is, the capacitors **C10** to **C40** are constantly charged from the capacitors **C11** to **C13** or the capacitors **C14** to **C16**. Therefore, even when electric currents flow from the nodes **N1** to **N4** to the output switching circuits **13A** and **13B** at a high speed, it is possible to suppress fluctuations in potentials of the nodes **N1** to **N4** because the nodes **N1** to **N4** are replenished with electric charges at a high speed.

[0120] By operating in this manner, the switched-capacitor circuit **12** can maintain an approximately equal voltage at both ends of each of the capacitors **C10**, **C20**, **C30**, and **C40**. Specifically, the voltages **V1** to **V4** (voltages relative to the ground potential) that satisfy $V1:V2:V3:V4=1:2:3:4$ are maintained at four nodes labelled **V1** to **V4**. Voltage levels of the voltages **V1** to **V4** correspond to a plurality of discrete voltage levels that can be supplied by the switched-capacitor circuit **12** to the output switching circuits **13A** and **13B**.

[0121] It is noted that a voltage ratio ($V1:V2:V3:V4$) is not limited to (1:2:3:4). For example, the voltage ratio ($V1:V2:V3:V4$) may be (1:2:4:8) in an exemplary aspect.

[0122] In addition, a configuration of the switched-capacitor circuit **12** illustrated in FIG. 4 is merely an example, and is not limited thereto. In FIG. 4, the switched-capacitor circuit **12** is configured to be able to supply four discrete voltages, but the configuration is not limited thereto. The switched-capacitor circuit **12** may be configured to be able to supply voltages of any number of discrete voltage levels, which is 2 or more. For example, in the case of supplying two discrete

voltages, the switched-capacitor circuit **12** should include at least the capacitors **C12** and **C15** as well as the switches **S21** to **S24** and **S31** to **S34**.

[2.2 Circuit Configuration of Output Switching Circuits **13A** and **13B**]

[0123] As illustrated in FIG. **4**, the output switching circuit **13A** includes input terminals **131A** to **134A**, switches **S51A** to **S54A**, and an output terminal **130A**.

[0124] The output terminal **130A** is connected to an input terminal **140A** of the filter circuit **14A**. The output terminal **130A** is a terminal for supplying a power supply voltage selected from the voltages **V1** to **V4** to the PA module **2A** via the filter circuit **14A**.

[0125] The input terminals **131A** to **134A** are connected to the nodes **N4** to **N1** of the switched-capacitor circuit **12**, respectively. The input terminals **131A** to **134A** are terminals for receiving the voltages **V4** to **V1** from the switched-capacitor circuit **12**.

[0126] The switch **S51A** is connected between the input terminal **131A** and the output terminal **130A**. Specifically, the switch **S51A** has a terminal connected to the input terminal **131A** and a terminal connected to the output terminal **130A**. In this connection configuration, by being switched ON and OFF by a control signal **S3A**, the switch **S51A** can switch between connection and disconnection between the input terminal **131A** and the output terminal **130A**.

[0127] The switch **S52A** is connected between an input terminal **132A** and the output terminal **130A**. Specifically, the switch **S52A** has a terminal connected to the input terminal **132A** and a terminal connected to the output terminal **130A**. In this connection configuration, by being switched ON and OFF by a control signal **S3A**, the switch **S52A** can switch between connection and disconnection between the input terminal **132A** and the output terminal **130A**.

[0128] The switch **S53A** is connected between an input terminal **133A** and the output terminal **130A**. Specifically, the switch **S53A** has a terminal connected to the input terminal **133A** and a terminal connected to the output terminal **130A**. In this connection configuration, by being switched ON and OFF by a control signal **S3A**, the switch **S53A** can switch between connection and disconnection between the input terminal **133A** and the output terminal **130A**.

[0129] The switch **S54A** is connected between the input terminal **134A** and the output terminal **130A**. Specifically, the switch **S54A** has a terminal connected to the input terminal **134A** and a terminal connected to the output terminal **130A**. In this connection configuration, by being switched ON and OFF by a control signal **S3A**, the switch **S54A** can switch between connection and disconnection between the input terminal **134A** and the output terminal **130A**.

[0130] The switches **S51A** and **S52A** are each at least one switch included in the output switching circuit **13A**.

[0131] These switches **S51A** to **S54A** are controlled to be exclusively ON. That is, only one of the switches **S51A** to **S54A** is turned ON, and remaining switches of the switches **S51A** to **S54A** are turned OFF. This allows the output switching circuit **13A** to output one voltage selected from the voltages **V1** to **V4**.

[0132] As illustrated in FIG. **4**, the output switching circuit **13B** includes input terminals **131B** to **134B**, switches **S51B** to **S54B**, and an output terminal **130B**.

[0133] The output terminal **130B** is connected to an input terminal **140B** of the filter circuit **14B**. The output terminal **130B** is a terminal for supplying a power supply voltage selected from the voltages **V1** to **V4** to the PA module **2B** via the filter circuit **14B**.

[0134] The input terminals **131B** to **134B** are connected to the nodes **N4** to **N1** of the switched-capacitor circuit **12**, respectively. The input terminals **131B** to **134B** are terminals for receiving the voltages **V4** to **V1** from the switched-capacitor circuit **12**.

[0135] Since configurations of the switches **S51B** and **S54B** are similar to those of the switches **S51A** to **S54A**, a description thereof is omitted here.

[0136] The switches **S51B** and **S52B** are each at least one switch included in the output switching circuit **13B**.

[0137] These switches **S51B** to **S54B** are controlled to be exclusively ON. That is, only one of the

switches **S51B** is turned ON, and remaining switches of the switches **S51B** to **S54B** are turned OFF. This allows the output switching circuit **13B** to output one voltage selected from the voltages **V1** to **V4**.

[0138] It is noted that configurations of the output switching circuits **13A** and **13B** illustrated in FIG. **4** are merely examples, and are not limited thereto. In particular, the switches **S51A** to **S54A** may have any configuration as long as they can selectively connect at least one of the four input terminals **131A** to **134A** to the output terminal **130A**. In addition, the switches **S51B** to **S54B** may have any configuration as long as they can selectively connect at least one of the four input terminals **131B** to **134B** to the output terminal **130B**. For example, the output switching circuit **13A** may further include a switch that is connected between the switches **S51A** to **S53A** and the switches **S54A** and the output terminal **130A**. In addition, for example, the output switching circuit **13B** may further include a switch connected between the switches **S51B** to **S53B** and the switch **S54B** and the output terminal **130B**. In addition, for example, the output switching circuit **13A** may further include a switch between the switches **S51A** and **S52A** and the switches **S53A** and **S54A** and the output terminal **130A**. In addition, for example, the output switching circuit **13B** may further include a switch between the switches **S51B** and **S52B** and the switches **S53B** and **S54B** and the output terminal **130B**.

[0139] It is noted that when two discrete voltages are supplied from the switched-capacitor circuit **12** according to an exemplary aspect, the output switching circuit **13A** may include at least two of the switches **S51A** to **S54A**, and the output switching circuit **13B** should include at least two of the switches **S51B** to **S54B**.

[2.3 Circuit Configuration of Pre-Regulator Circuit **11**]

[0140] As illustrated in FIG. **4**, the pre-regulator circuit **11** includes an input terminal **110**, output terminals **111** to **114**, inductor connection terminals **115** and **116**, switches **S61** to **S63**, as well as **S71** and **S72**, a power inductor **L71**, and capacitors **C61** to **C64**.

[0141] The input terminal **110** is a direct current voltage input terminal for receiving an input voltage from a direct-current power supply.

[0142] The output terminal **111** is an output terminal for the voltage **V4**. That is, the output terminal **111** is a terminal for supplying the voltage **V4** to the switched-capacitor circuit **12**. The output terminal **111** is connected to the node **N4** of the switched-capacitor circuit **12**.

[0143] The output terminal **112** is an output terminal for the voltage **V3**. That is, the output terminal **112** is a terminal for supplying the voltage **V3** to the switched-capacitor circuit **12**. The output terminal **112** is connected to the node **N3** of the switched-capacitor circuit **12**.

[0144] The output terminal **113** is an output terminal for the voltage **V2**. That is, the output terminal **113** is a terminal for supplying the voltage **V2** to the switched-capacitor circuit **12**. The output terminal **113** is connected to the node **N2** of the switched-capacitor circuit **12**.

[0145] The output terminal **114** is an output terminal for the voltage **V1**. That is, the output terminal **114** is a terminal for supplying the voltage **V1** to the switched-capacitor circuit **12**. The output terminal **114** is connected to the node **N1** of the switched-capacitor circuit **12**.

[0146] The inductor connection terminal **115** is connected to one end of the power inductor **L71**. The inductor connection terminal **116** is connected to another end of the power inductor **L71**.

[0147] The switch **S71** is connected between the input terminal **110** and the one end of the power inductor **L71**. Specifically, the switch **S71** has a terminal connected to the input terminal **110** and a terminal connected to the one end of the power inductor **L71** via the inductor connection terminal **115**. In this connection configuration, by switching ON/OFF based on a control signal **S1**, the switch **S71** can switch between connection and disconnection between the input terminal **110** and the one end of the power inductor **L71**.

[0148] The switch **S72** is connected between the one end of the power inductor **L71** and the ground. Specifically, the switch **S72** has a terminal connected to the one end of the power inductor **L71** via the inductor connection terminal **115**, and a terminal connected to the ground. In this

connection configuration, by switching ON/OFF based on the control signal **S1**, the switch **S72** can switch between connection and disconnection between the one end of the power inductor **L71** and the ground.

[0149] The switch **S61** is connected between the other end of the power inductor **L71** and the output terminal **111**. Specifically, the switch **S61** has a terminal connected to the other end of the power inductor **L71** via the inductor connection terminal **116**, and a terminal connected to the output terminal **111**. In this connection configuration, by switching ON/OFF based on the control signal **S1**, the switch **S61** can switch between connection and disconnection between the other end of the power inductor **L71** and the output terminal **111**.

[0150] A switch **S62** is connected between the other end of the power inductor **L71** and the output terminal **112**. Specifically, the switch **S62** has a terminal connected to the other end of the power inductor **L71** via the inductor connection terminal **116**, and a terminal connected to the output terminal **112**. In this connection configuration, by switching ON/OFF based on the control signal **S1**, the switch **S62** can switch between connection and disconnection between the other end of the power inductor **L71** and the output terminal **112**.

[0151] The switch **S63** is connected between the other end of the power inductor **L71** and the output terminal **113**. Specifically, the switch **S63** has a terminal connected to the other end of the power inductor **L71** via the inductor connection terminal **116**, and a terminal connected to the output terminal **113**. In this connection configuration, by switching ON/OFF based on the control signal **S1**, the switch **S63** can switch between connection and disconnection between the other end of the power inductor **L71** and the output terminal **113**.

[0152] The switches **S71** and **S72** are each at least one switch included in the pre-regulator circuit **11**.

[0153] One of two electrodes of the capacitor **C61** is connected to the switch **S61** and the output terminal **111**. The other of the two electrodes of the capacitor **C61** is connected to the switch **S62**, the output terminal **112**, and one of two electrodes of the capacitor **C62**. The capacitor **C61** is configured to function as a smoothing capacitor in an exemplary aspect.

[0154] The one of the two electrodes of the capacitor **C62** is connected to the switch **S62**, the output terminal **112**, and the other of the two electrodes of the capacitor **C61**. The other of the two electrodes of the capacitor **C62** is connected to the switch **S63**, the output terminal **113**, and one of the two electrodes of the capacitor **C63**.

[0155] The one of the two electrodes of the capacitor **C63** is connected to the switch **S63**, the output terminal **113**, and the other of the two electrodes of the capacitor **C62**. The other of the two electrodes of the capacitor **C63** is connected to the output terminal **114** and one of the two electrodes of the capacitor **C64**.

[0156] The one of the two electrodes of the capacitor **C64** is connected to the output terminal **114** and the other of the two electrodes of the capacitor **C63**. The other of the two electrodes of the capacitor **C64** is connected to the ground.

[0157] The switches **S61** to **S63** are controlled to be exclusively ON. That is, only one of the switches **S61** to **S63** is turned ON, and remaining switches of the switches **S61** to **S63** are turned OFF. By turning ON only one of the switches **S61** to **S63**, the pre-regulator circuit **11** can change a voltage to be supplied to the switched-capacitor circuit **12** at the voltage levels of the voltages **V2** to **V4**.

[0158] The pre-regulator circuit **11** configured in this manner can supply electric charges to the switched-capacitor circuit **12** via at least one of the output terminals **111** to **113**.

[0159] It is noted that when an input voltage to the pre-regulator circuit **11** is converted to one voltage according to an exemplary aspect, the pre-regulator circuit **11** should include at least the switches **S71** and **S72**, and the power inductor **L71**.

[2.4 Circuit Configuration of Filter Circuits **14A** and **14B**]

[0160] As illustrated in FIG. **4**, the filter circuit **14A** includes inductors **L1** and **L2**, a capacitor **C1**,

a switch **SW1**, the input terminal **140A**, and the external output terminal **101**.

[0161] The input terminal **140A** is connected to the output terminal **130A** of the output switching circuit **13A**. The input terminal **140A** is a terminal for receiving a voltage selected from a plurality of discrete voltages by the output switching circuit **13A**.

[0162] The external output terminal **101** is an external connection terminal of the ET module **1A** and is connected to the PA module **2A**. The external output terminal **101** is a terminal for supplying the PA module **2A** with the power supply voltage $V_{sub.ETA}$ which is a plurality of discrete voltages that have passed through the filter circuit **14A**.

[0163] The inductor **L1** is connected between the input terminal **140A** and the external output terminal **101**. That is, the inductor **L1** is disposed in series in a path connecting the input terminal **140A** and the external output terminal **101**. Specifically, one end of the inductor **L1** is connected to the input terminal **140A**, and another end of the inductor **L1** is connected to the external output terminal **101**.

[0164] The inductor **L2** is connected between a path connecting between the inductor **L1** and the external output terminal **101**, and the ground. That is, the inductor **L2** is shunt-connected to the path connecting the input terminal **140A** and the external output terminal **101**. Specifically, one end of the inductor **L2** is connected to a node **N42** on the path connecting the inductor **L1** and the external output terminal **101**, and another end of the inductor **L2** is connected to the ground via the capacitor **C1**. It is noted that the inductor **L2** may be connected between the capacitor **C1** and the ground, and may be omitted from the filter circuit **14A** in an exemplary aspect.

[0165] The capacitor **C1** is connected between the inductor **L2** and the ground. That is, the capacitor **C1** is shunt-connected to the path connecting the input terminal **140A** and the external output terminal **101**. Specifically, one end of the capacitor **C1** is connected to the inductor **L2** and another end of the capacitor **C1** is connected to the ground.

[0166] The switch **SW1** is connected between the input terminal **140A** and the external output terminal **101** not via the inductor **L1**. That is, the switch **SW1** is disposed in series in a path that bypasses the inductor **L1** between the input terminal **140A** and the external output terminal **101**. Specifically, one end of the switch **SW1** is connected to a node **N41** on a path connecting the input terminal **140A** and the inductor **L1**, and another end of the switch **SW1** is connected to a node **N43** on the path connecting the inductor **L1** and the external output terminal **101**.

[0167] It is noted that in FIG. 4, the switch **SW1** and the inductor **L1** are connected in parallel, but another circuit element may be inserted in a path of the switch **SW1** and/or a path of the inductor **L1**. For example, an inductor may be connected between the switch **SW1** and the node **N43** and/or between the switch **SW1** and the node **N41**.

[0168] In addition, in FIG. 4, although the node **N43** to which the other end of the switch **SW1** is connected is located between the node **N42** to which the inductor **L2** is connected and the external output terminal **101**, a positional relationship between the node **N42** and the node **N43** is not limited thereto. For example, the node **N42** may be located between the node **N43** and the external output terminal **101**. In addition, for example, a position of the node **N42** may be the same as that of the node **N43**.

[0169] The switch **SW1** connected in this manner is switched ON/OFF based on a control signal **S4A**. This allows the filter circuit **14A** to switch ON/OFF a band-stop filter for removing noise from a plurality of discrete voltages.

[0170] ON/OFF of such a band-stop filter can be controlled based on, for example, a channel band width (i.e., modulation band width) of a radio frequency signal. In addition, when the PA module **2A** can amplify transmission signals of a plurality of frequency bands, the ON/OFF of the switch **SW1** may be controlled based on the frequency bands of the transmission signals amplified by the PA module **2A**. It is noted that control of the band-stop filter is not limited to the above as would be appreciated to one skilled in the art.

[0171] As illustrated in FIG. 4, the filter circuit **14B** includes inductors **L3** and **L4**, a capacitor **C2**,

a switch SW2, the input terminal **140B**, and the external output terminal **104**. A configuration of the filter circuit **14B** is described by replacing the inductors L1 and L2 in the filter circuit **14A** with the inductors L3 and L4, replacing the capacitor C1 in the filter circuit **14A** with the capacitor C2, replacing the switch SW1 in the filter circuit **14A** with the switch SW2, replacing the input terminal **140A** in the filter circuit **14A** with the input terminal **140B**, replacing the external output terminal **101** in the filter circuit **14A** with the external output terminal **104**, and replacing the control signal S4A with a control signal S4B.

[0172] It is noted that the filter circuits **14A** and **14B** may be low pass filters for removing noise from the plurality of discrete voltages according to an exemplary aspect.

[3. Circuit Configuration of ET Module **1B** and Communication Device **5B** in Example 2]

[0173] FIG. 5 is a circuit configuration diagram of an ET module **1B** and a communication device **5B** according to Example 2. The ET module **1B** and the communication device **5B** according to this Example are examples of the ET module **1** and the communication device **5** according to the present embodiment, and illustrate a specific example of the circuit configuration of the ET module **1**.

[0174] The communication device **5B** according to this Example includes the ET module **1B**, the PA module **2B**, the antennae **3A** and **3B**, and the RFIC **4**. The communication device **5B** according to this Example differs from the communication device **5** according to the embodiment in the configuration of the ET module **1B**. Therefore, in the following, the communication device **5B** according to this Example is mainly described with a focus on the configuration of the ET module **1B**.

[0175] The ET module **1B** is an example of a tracker module and can supply the power supply voltages V.sub.ETA and V.sub.ETB, the voltage levels of which vary, to the PA modules **2A** and **2B**, respectively, based on a tracking mode. The analog ET mode and the APT mode can be used as a tracking mode, but the tracking mode is not limited thereto. As illustrated in FIG. 5, the ET module **1B** includes a voltage supply circuit **10B**, the control circuit **70**, the external output terminals **101** and **104**, the external input terminals **102** and **105**, and the control signal terminal **103**. The ET module **1B** according to this Example differs from the ET module **1** according to the embodiment in a configuration of the voltage supply circuit **10B**. In the following, the ET module **1B** according to this Example is described, omitting the same configurations as those of the ET module **1** according to the embodiment and focusing on different configurations.

[0176] The voltage supply circuit **10B** includes DC-DC converters **15A** and **15B** and linear amplifier circuits **16A** and **16B**.

[0177] Each of the DC-DC converters **15A** and **15B** has, for example, one power inductor and outputs a direct-current voltage. It is noted that based on a control signal received from the control circuit **70**, the DC-DC converters **15A** and **15B** can change a level of the direct current voltage for each symbol or for average electric power corresponding to the DC-DC converters **15A** and **15B**.

[0178] Each of the linear amplifier circuits **16A** and **16B** receives, from the RFIC **4**, an envelope signal representative of a power supply voltage waveform based on an envelope value, and linearly amplifies the envelope signal.

[0179] In operation, when the analog ET mode is applied to the ET module **1i**, for example, the linear amplifier circuits **16A** and **16B** are operated, and the power supply voltages V.sub.ETA and V.sub.ETB, which correspond to an envelope value and the voltage levels of which vary, are generated at the voltage supply circuit **10B**. The voltage supply voltages V.sub.ETA is supplied to the PA module **2A** via the external output terminal **101**, and the voltage supply voltages V.sub.ETB is supplied to the PA module **2B** via the external output terminal **104**. In addition, when the APT mode is applied to the ET module **1B**, for example, the linear amplifier circuits **16A** and **16B** are not operated, and the power supply voltages V.sub.ETA and V.sub.ETB, which are stabilized, are generated at the voltage supply circuit **10B**. The voltage supply voltages V.sub.ETA is supplied to the PA module **2A** via the external output terminal **101**, and the voltage supply voltages V.sub.ETB

is supplied to the PA module **2B** via the external output terminal **104**.

[4. Implementation Examples of ET Module **1A** According to Example 1]

[0180] Next, as an implementation example of the ET module **1** configured as described above, an implementation configuration of the ET module **1A** according to Example 1 is described with reference to FIGS. **6A** to **6C**.

[0181] It is noted that in this Example, the power inductor **L71** included in the pre-regulator circuit **11** is not disposed on a module laminate **90**, but is not limited thereto. That is, the power inductor **L71** may be disposed on the module laminate **90**.

[0182] FIG. **6A** is a plan view of the ET module **1A** according to Example 1. FIG. **6B** is a plan view of the ET module **1A** according to Example 1. FIG. **6A** is a view of a main surface **90a** side of the module laminate **90** viewed from the z-axis positive side. FIG. **6B** is a perspective view of a main surface **90b** side of the module laminate **90** from the z-axis positive side. FIG. **6C** a cross-sectional view of the ET module **1A** according to Example 1. A cross section of the ET module **1A** in FIG. **6C** is a cross section taken along VIC-VIC line in FIGS. **6A** and **6B**.

[0183] It is noted that in FIGS. **6A** to **6C**, illustration of some of wiring lines connecting a plurality of circuit components disposed on the module laminate **90** is omitted. In FIGS. **6A** and **6B**, illustration of a resin member **91** covering the plurality of circuit components is omitted.

Furthermore, in FIG. **6C**, illustration of a shield electrode layer covering the resin member **91** is omitted. It is also noted that the resin member **91** and the shield electrode layer may be omitted in an exemplary aspect. In addition, it is noted that hatched blocks in FIG. **6A** represent optional circuit components that may be omitted in exemplary aspects.

[0184] As illustrated in FIGS. **6A** and **6B**, the ET module **1A** includes the module laminate **90**, an integrated circuit **80**, the external output terminals **101** and **104**, and the external input terminals **102** and **105**.

[0185] The module laminate **90** has the main surfaces **90a** and **90b** that face each other. A ground plane or the like is formed in the module laminate **90** and on the main surface **90a**. It is noted that in FIGS. **6A** and **6B**, the module laminate **90** has a rectangular shape in plan view, but the shape of the module laminate **90** is not limited thereto as would be appreciated to one skilled in the art.

[0186] The module laminate **90** may be, but is not limited to, for example, a low temperature co-fired ceramics (LTCC: Low Temperature Co-fired Ceramics) board or a high temperature co-fired ceramics (HTCC: High Temperature Co-fired Ceramics) board, which have a laminated structure of a plurality of dielectric layers, a component-embedded board, a board having a redistribution layer (RDL: Redistribution Layer), or a printed circuit board, or the like.

[0187] The module laminate **90** has sides **P1** (a first side), **P2** (a second side), **P3**, and **P4** that form at least a part of an outer shape of the module laminate **90** in plan view of the module laminate **90**. The side **P1** is adjacent to the sides **P2** and **P4**. The side **P2** is adjacent to the sides **P1** and **P3**. The side **P3** is adjacent to the sides **P2** and **P4**. The side **P4** is adjacent to the sides **P1** and **P3**.

[0188] The integrated circuit **80** is disposed on the main surface **90a** of the module laminate **90**, and has a PR switch portion **11S**, an SC switch portion **12S**, an OS switch portion **13S**, an FL switch portion **14S**, and a digital control section **70A**. The PR switch portion **11S** includes the switches **S61** to **S63**, **S71**, and **S72** of the pre-regulator circuit **11**. The SC switch portion **12S** includes the switches **S11** to **S14**, **S21** to **S24**, **S31** to **S34**, and **S41** to **S44** of the switched-capacitor circuit **12**. The OS switch portion **13S** includes the switches **S51A** to **S54A** of the output switching circuit **13A** and the switches **S51B** to **S54B** of the output switching circuit **13B**. The FL switch portion **14S** includes the switch **SW1** of the filter circuit **14A** and the switch **SW2** of the filter circuit **14B**. The digital control section **70A** includes the control circuit **70**.

[0189] In addition, in FIG. **6A**, the integrated circuit **80** has a rectangular shape in plan view of the module laminate **90**, but the shape of the integrated circuit **80** is not limited thereto.

[0190] The integrated circuit **80** includes, for example, a CMOS (Complementary Metal Oxide Semiconductor), and, specifically, may be manufactured by an SOI (Silicon on Insulator) process.

It is noted that the integrated circuit **80** is not limited to a CMOS.

[0191] It is noted Note that the integrated circuit **80** is not necessarily configured by, for example, a single chip, and may be configured by a plurality of chips. In addition, the integrated circuit **80** only should include at least the SC switch portion **12S** and the OS switch portion **13S**, and the PR switch portion **11S**, the FL switch portion **14S**, and the digital control section **70A** may be disposed outside of the integrated circuit **80**.

[0192] Furthermore, as illustrated in FIG. **6B**, a plurality of external connection terminals **150** is disposed on the main surface **90B**. The plurality of external connection terminals **150** is electrically connected to a plurality of electronic components disposed on the main surface **90a** via a via conductor or the like formed in the module laminate **90**. The plurality of external connection terminals **150** may be a copper electrode but is not limited thereto. For example, a solder electrode may be used as the plurality of external connection terminals **150**.

[0193] Each of the external output terminals **101** and **104**, and the external input terminals **102** and **105** is one of the plurality of external connection terminals **150**. Some of the plurality of external connection terminals **150** other than the external output terminals **101** and **104**, and the external input terminals **102** and **105** are set to the ground.

[0194] Here, as illustrated in FIG. **6B**, the external output terminal **101** is disposed adjacent to the side **P1**, and the external output terminal **104** is not disposed adjacent to the side **P1**, but is disposed adjacent to the side **P2**.

[0195] As a result of this, as compared to the configuration in which the external output terminals **101** and **104** are adjacently disposed on the same side of the module laminate **90**, the wiring lines supplying the power supply voltages $V_{sub.ETA}$ and $V_{sub.ETB}$ can be disposed at a distance from each other. Therefore, the heat dissipation of the ET module **1A** can be improved, and the waveform distortion of the power supply voltages $V_{sub.ETA}$ and $V_{sub.ETB}$ caused by the heat generation can be suppressed, thereby suppressing the efficiency deterioration.

[0196] In addition, the external input terminal **102** is disposed adjacent to the side **P1**. As a result of this, the external input terminal **102** and the external output terminal **101** are disposed in proximity, so that the total wiring length of the wiring line connecting the external output terminal **101** and the PA module **2A** and the wiring line connecting the external input terminal **102** and the PA module **2A** can be reduced. Therefore, the waveform distortion of the power supply voltage $V_{sub.ETA}$ output from the ET module **1A** can be suppressed, thereby suppressing the efficiency deterioration.

[0197] Furthermore, in the plan view of the module laminate **90**, the external output terminal **101** and the external input terminal **102** are adjacently disposed. This configuration further reduces the above total wiring length.

[0198] In addition, the external input terminal **105** is disposed adjacent to the side **P2**. As a result of this, the external input terminal **105** and the external output terminal **104** are disposed in proximity, so that the total wiring length of the wiring line connecting the external output terminal **104** and the PA module **2B** and the wiring line connecting the external input terminal **105** and the PA module **2B** can be reduced. Therefore, the waveform distortion of the power supply voltage $V_{sub.ETB}$ output from the ET module **1A** can be suppressed, thereby suppressing the efficiency deterioration.

[0199] Furthermore, in the plan view of the module laminate **90**, the external output terminal **104** and the external input terminal **105** are adjacently disposed. This configuration further reduces the above total wiring length.

[0200] In addition, as illustrated in FIG. **6B**, in the plan view of the module laminate **90**, the external connection terminal **150** (GND) set to the ground potential is disposed adjacent to the external output terminal **101**, and the external connection terminal **150** (GND) set to the ground potential is disposed adjacent to the external output terminal **104**.

[0201] As a result of this, interference between the power supply voltage $V_{sub.ETA}$ supplied via the external output terminal **101** and the power supply voltage $V_{sub.ETB}$ supplied via the external output terminal **104** can be suppressed, thereby suppressing the waveform distortion of the power

supply voltage V.sub.ETA and the power supply voltage V.sub.ETB.

[0202] It is noted that in the plan view of the module laminate **90**, the external connection terminal **150** (GND) set to the ground potential may be disposed between the external output terminal **101** and the external output terminal **104**.

[0203] As a result of this, the interference between the power supply voltage V.sub.ETA supplied via the external output terminal **101** and the power supply voltage V.sub.ETB supplied via the external output terminal **104** can be suppressed, thereby suppressing the waveform distortion of the power supply voltage V.sub.ETA and the power supply voltage V.sub.ETB.

[0204] Furthermore, in addition to the ET module **1A** includes the module laminate **90**, the integrated circuit **80**, and a plurality of the external connection terminals **150**, the ET module **1A** includes the capacitors **C61** to **C64** included in the pre-regulator circuit **11**, the capacitors **C11** to **C16** and the capacitors **C10** to **C40** included in the switched-capacitor circuit **12**, the inductors **L1** and **L2** as well as the capacitor **C1** included in the filter circuit **14A**, the inductors **L3** and **L4** as well as the capacitor **C2** included in the filter circuit **14B**, and the resin member **91**.

[0205] The integrated circuit **80**, the capacitors **C61** to **C64**, the capacitors **C11** to **C16**, the capacitors **C10** to **C40**, the inductors **L1** to **L4** as well as the capacitors **C1** and **C2**, and the resin member **91** are disposed on the main surface **90a**.

[0206] The resin member **91** covers at least some of the plurality of electronic components on the main surface **90a**. The resin member **91** can be configured to ensure reliability such as mechanical strength and moisture resistance or the like of the plurality of electronic components on the main surface **90a**. It is noted that the resin member **91** may be omitted from the ET module **1A** in an exemplary aspect.

[0207] Each of the capacitors **C61** to **C64**, the capacitors **C11** to **C16**, and the capacitors **C10** to **C40** is implemented as a chip capacitor. In this aspect, a chip capacitor can refer to a surface mount device (SMD: Surface Mount Device) that forms a capacitor. It is noted that the implementation of a plurality of the capacitors is not limited to a chip capacitor. For example, some or all of the plurality of capacitors may be included in an integrated passive device (IPD: Integrated Passive Device) or may be included in the integrated circuit **80** in various exemplary aspects.

[0208] The inductors **L1** to **L4** are implemented as chip inductors. A chip inductor can refer to an SMD that forms an inductor. It is noted that the implementation of the inductors **L1** to **L4** is not limited to a chip inductor. For example, some or all of the inductors **L1** to **L4** may be included in an IPD in various exemplary aspects.

[0209] The plurality of capacitors and inductors disposed in this manner on the main surface **90a** is grouped for each circuit and is disposed around the integrated circuit **80**.

[0210] Specifically, a group of the capacitors **C61** to **C64** included in the pre-regulator circuit **11** is disposed in a region on the main surface **90a** sandwiched between a straight line along a left edge of the integrated circuit **80** and a straight line along a left edge of the module laminate **90** in plan view of the module laminate **90**. As a result, a group of circuit components included in the pre-regulator circuit **11** is disposed close to the PR switch portion **11S** in the integrated circuit **80**.

[0211] In plan view of the module laminate **90**, a group of the capacitors **C11** to **C16** and the capacitors **C10** to **C40** included in the switched-capacitor circuit **12** is disposed in a region on the main surface **90a** sandwiched between a straight line along an upper edge of the integrated circuit **80** and a straight line along an upper edge of the module laminate **90**, as well as in a region on the main surface **90a** sandwiched between a straight line along a right edge of the integrated circuit **80** and a straight line along a right edge of the module laminate **90**. As a result, a group of circuit components included in the switched-capacitor circuit **12** is disposed close to the SC switch portion **12S** of the integrated circuit **80**. That is, the SC switch portion **12S** is disposed closer to the capacitors **C11** to **C16** and the capacitors **C10** to **C40** of the switched-capacitor circuit **12** than each of the PR switch portion **11S** and the OS switch portion **13S** is.

[0212] In plan view of the module laminate **90**, a group of the inductors **L1** to **L4** and the capacitors

C1 and C2 included in the filter circuits 14A and 14B is disposed in a region on the main surface 90a sandwiched between a straight line along a lower edge of the integrated circuit 80 and a straight line along a lower edge of the module laminate 90. As a result, a group of circuit components included in the filter circuits 14A and 14B is disposed close to the FL switch portion 14S in the integrated circuit 80. That is, the FL switch portion 14S is disposed closer to the inductors L1 to L4 and the capacitors C1 and C2 of the filter circuits 14A and 14B than each of the PR switch portion 11S and the SC switch portion 12S is.

[0213] It is noted that configurations of the ET module 1A illustrated in FIGS. 6A to 6C are merely examples, and are not limited thereto. For example, some of the capacitors and the inductors disposed on the main surface 90a may be formed within the module laminate 90. In addition, some of the capacitors and the inductors disposed on the main surface 90a may be omitted from the ET module 1A and are not disposed on the module laminate 90.

[5. Arrangement of ET Module 1A and Communication Device 5A According to Example 1]

[0214] Next, a description is given of the arrangement on a motherboard 92 of the communication device 5A according to Example 1 with reference to FIG. 7. FIG. 7 is a plan view of the communication device 5A according to Example 1. As illustrated in FIG. 7, the communication device 5A includes the motherboard 92, the RFIC 4, the ET module 1A, the PA modules 2A and 2B, and the capacitors 31A and 31B. The PA module 2A includes the power amplifiers 41 to 43, and the PA module 2B includes the power amplifiers 44 to 46.

[0215] The motherboard 92 can be, but is not limited to, for example, a printed circuit board, an LTCC board or an HTCC board, which have a laminated structure of a plurality of dielectric layers, a component-embedded board, a board having an RDL, or the like.

[0216] The RFIC 4, the ET module 1A, the PA modules 2A and 2B, and the capacitors 31A and 31B are disposed on a main surface of the motherboard 92.

[0217] The capacitor 31A is disposed on the motherboard 92 between the ET module 1A and the PA module 2A. The capacitor 31A is connected between a node on the first path connecting the external output terminal 101 and the PA module 2A, and the external input terminal 102. The capacitor 31A is implemented as a chip capacitor. It is noted that the capacitor 31A is not limited to a chip capacitor, and may be included in an IPD, or may be formed of a planar electrode in the motherboard 92 in various exemplary aspects.

[0218] The external output terminal 101 is disposed adjacent to the side P1, and the external output terminal 104 is not disposed adjacent to the side P1, but is disposed adjacent to the side P2.

[0219] As a result of this, as compared to the configuration in which the external output terminals 101 and 104 are adjacently disposed on the same side of the module laminate 90, the wiring lines supplying the power supply voltages V.sub.ETA and V.sub.ETB can be disposed at a distance from each other. Therefore, the heat dissipation of the ET module 1A can be improved, and the waveform distortion of the power supply voltages V.sub.ETA and V.sub.ETB caused by the heat generation can be suppressed, thereby suppressing the efficiency deterioration.

[0220] In addition, the external input terminal 102 is disposed adjacent to the side P1. As a result of this, the external input terminal 102 and the external output terminal 101 are disposed in proximity, so that the total wiring length of the wiring line connecting the external output terminal 101 and the PA module 2A and the wiring line connecting the external input terminal 102 and the PA module 2A can be reduced. Therefore, the waveform distortion of the power supply voltage V.sub.ETA output from the ET module 1A can be suppressed, thereby suppressing the efficiency deterioration.

[0221] Furthermore, in the plan view of the module laminate 90, the external output terminal 101 and the external input terminal 102 are adjacently disposed. This configuration further reduces the above total wiring length.

[0222] In addition, the external input terminal 105 is disposed adjacent to the side P2. As a result of this, the external input terminal 105 and the external output terminal 104 are disposed in proximity, so that the total wiring length of the wiring line connecting the external output terminal 104 and the

PA module **2B** and the wiring line connecting the external input terminal **105** and the PA module **2B** can be reduced. Therefore, the waveform distortion of the power supply voltage $V_{sub.ETB}$ output from the ET module **1A** can be suppressed, thereby suppressing the efficiency deterioration. [0223] Furthermore, in the plan view of the module laminate **90**, the external output terminal **104** and the external input terminal **105** are adjacently disposed. This configuration further reduces the above total wiring length.

[0224] In addition, the external output terminal **101** and the external input terminal **102** are connected to the capacitor **31A**. As a result of this configuration, when the APT mode is applied to the ET module **1A**, the capacitor **31A** can be configured to function as the bypass condenser, and the voltage level of the power supply voltage $V_{sub.ETA}$ can be stabilized.

[0225] In addition, the external output terminal **104** and the external input terminal **105** are connected to the capacitor **31B**. As a result of this configuration, when the APT mode is applied to the ET module **1A**, the capacitor **31B** can be configured to function as the bypass condenser, and the voltage level of the power supply voltage $V_{sub.ETB}$ can be stabilized.

[6. Arrangement of ET module **1B** and Communication Device **5B** According to Example 2]

[0226] Next, a description is given of the arrangement on the motherboard **92** of the communication device **5B** according to Example 2 with reference to FIG. **8**. FIG. **8** is a plan view of the communication device **5B** according to Example 2. As illustrated in FIG. **8**, the communication device **5B** includes the motherboard **92**, the RFIC **4**, the ET module **1i**, the PA modules **2A** and **2B**, and inductors **33A** and **33B**. The PA module **2A** includes the power amplifiers **41** to **43**, and the PA module **2B** includes the power amplifiers **44** to **46**. It is noted that as compared to the circuit configuration of the communication device **5B** illustrated in FIG. **5**, the inductors **33A** and **33B** are added to the communication device **5B** illustrated in FIG. **8**. An implementation configuration of the communication device **5B** illustrated in FIG. **8** mainly differs from that of the communication device **5A** illustrated in FIG. **7** in that inductors **33A** and **33B** are disposed instead of the capacitors **31A** and **31B**. In the following, the implementation configuration of the communication device **5B** according to Example 2 is described with a focus on different points from the implementation configuration of the communication device **5A** according to Example 1, and a description of the same points is omitted.

[0227] The RFIC **4**, the ET module **1B**, the PA modules **2A** and **2B**, and the inductors **33A** and **33B** are disposed on the main surface of the motherboard **92**.

[0228] The inductor **33A** is disposed on the motherboard **92** between the ET module **1B** and the PA module **2A**. The inductor **33A** is disposed in series on the first path connecting the external output terminal **101** and the PA module **2A**.

[0229] As a result of this configuration, when the analog ET mode is applied to the ET module **1B**, the inductor **33A** can be configured to function as a choke coil that blocks a radio frequency signal leaking from the PA module **2A**. Therefore, superimposition of radio frequency noise on the power supply voltage $V_{sub.ETA}$ output from the external output terminal **101** can be suppressed.

[0230] The inductor **33B** is disposed on the motherboard **92** between the ET module **1B** and the PA module **2B**. The inductor **33B** is disposed in series on the second path connecting the external output terminal **104** and the PA module **2B**.

[0231] As a result of this configuration, when the analog ET mode is applied to the ET module **1B**, the inductor **33B** can be configured to function as a choke coil that blocks a radio frequency signal leaking from the PA module **2B**. Therefore, superimposition of radio frequency noise on the power supply voltage $V_{sub.ETB}$ output from the external output terminal **104** can be suppressed.

[0232] It is noted that each of the inductor **33A** and **33B** is implemented as a chip inductor. However, the inductors **33A** and **33B** are not limited to chip inductors and may be included in an IPD or may be formed of a planar coil in the motherboard **92** in various exemplary aspects.

[0233] The external input terminal **102** is connected to the voltage supply circuit **10B**. This configuration makes it possible to return a feedback signal of the power supply voltage $V_{sub.ETA}$

on the first path to the voltage supply circuit **10B** via the external input terminal **102**. In addition, the external input terminal **105** is connected to the voltage supply circuit **10B**. This makes it possible to return a feedback signal of the power supply voltage $V_{sub.ETB}$ on the second path to the voltage supply circuit **10B** via the external input terminal **105**. It is noted that the external input terminals **102** and **105** may be connected to the control circuit **70**, rather than being connected to the voltage supply circuit **10B**.

[0234] It is noted that the inductor **33A** may be connected between the external input terminal **102** and a node on the first path. As a result of this configuration, when the analog ET mode is applied to the ET module **1**, the inductor **33A** can be configured to function as the choke coil that blocks a radio frequency signal leaking from the PA module **2A**. Therefore, it is possible to return the feedback signal of the power supply voltage $V_{sub.ETA}$ on the first path to the ET module **1B** with high precision via the external input terminal **102**.

[0235] In addition, the inductor **33B** may be connected between the external input terminal **105** and a node on the second path. As a result of this configuration, when the analog ET mode is applied to the ET module **1B**, the inductor **33B** can be configured to function as the choke coil that blocks a radio frequency signal leaking from the PA module **2B**. Therefore, it is possible to return the feedback signal of the power supply voltage $V_{sub.ETB}$ on the second path to the ET module **1B** with high precision via the external input terminal **105**.

[7. Arrangement of ET Module **1C** and Communication Device **5C** According to Modification Example]

[0236] Next, a description is given of arrangement on a motherboard **92** of a communication device **5C** according to a modification example with reference to FIG. **9**. FIG. **9** is a plan view of the communication device **5C** according to the modification example. As illustrated in FIG. **9**, the communication device **5C** includes the motherboard **92**, the RFIC **4**, an ET module **1C**, the PA modules **2A**, **2B**, and **2C**, and the capacitors **31A** and **31B**.

[0237] The ET module **1C** includes the module laminate **90**, a voltage supply circuit **10C**, the control circuit **70**, the external output terminals **101**, **104**, and **106**, and the external input terminals **102** and **105**, and an external input terminal **107**.

[0238] The PA module **2A** includes the power amplifiers **41** to **43** and can amplify radio frequency signals in band A to and C that belong to a low band group (600 MHz to 1 GHz). The PA module **2B** includes the power amplifiers **44** to **46** and can amplify radio frequency signals in band D to band F that belong to a middle and high band groups (1.5 to 2.8 GHz). The PA module **2C** can amplify radio frequency signals in bands that belong to an ultra-high band group (3.3 to 5.0 GHz). The communication device **5C** according to this modification example mainly differs from the communication device **5A** according to Example 1 in that the PA module **2C** is added. In the following, an implementation configuration of the communication device **5C** according to this modification example is described with a focus on different points from the implementation configuration of the communication device **5A** according to Example 1, and a description of the same points is omitted.

[0239] The RFIC **4**, the ET module **1C**, the PA modules **2A**, **2B**, and **2C**, and the capacitors **31A** and **31B** are disposed on a main surface of the motherboard **92**.

[0240] The capacitor **31A** is disposed on the motherboard **92** between the ET module **1C** and the PA module **2A**. The capacitor **31A** is connected between a node on the first path connecting the external output terminal **101** and the PA module **2A**, and the external input terminal **102**.

[0241] The capacitor **31B** is disposed on the motherboard **92** between the ET module **1C** and the PA module **2B**. The capacitor **31B** is connected between a node on the second path connecting the external output terminal **104** and the PA module **2B**, and the external input terminal **105**.

[0242] The external output terminal **101** is disposed adjacent to the side **P1**. The external output terminal **104** is not disposed adjacent to the side **P1** but is disposed adjacent to the side **P2**. The external output terminal **106** is not disposed adjacent to the sides **P1** and **P2** but is disposed

adjacent to the side P3.

[0243] As result of this, as compared to a configuration in which the external output terminals **101**, **104**, and **106** are adjacently disposed on a same side of the module laminate **90**, wiring lines supplying the power supply voltage V.sub.ET can be disposed at a distance from each other. Therefore, heat dissipation of the ET module **1C** can be improved, and waveform distortion of the power supply voltages V.sub.ET caused by the heat generation can be suppressed, thereby suppressing efficiency deterioration.

[0244] In addition, the external input terminal **102** is disposed adjacent to the side P1. As a result of this, the external input terminal **102** and the external output terminal **101** are adjacently disposed in proximity, so that the total wiring length of the wiring line connecting the external output terminal **101** and the PA module **2A** and the wiring line connecting the external input terminal **102** and the PA module **2A** can be reduced. Therefore, the waveform distortion of the power supply voltage V.sub.ET output from the ET module **1C** can be suppressed, thereby suppressing the efficiency deterioration.

[0245] Furthermore, in the plan view of the module laminate **90**, the external output terminal **101** and the external input terminal **102** are adjacently disposed. This configuration further reduces the above total wiring length.

[0246] In addition, the external input terminal **105** is disposed adjacent to the side P2. As a result of this, the external input terminal **105** and the external output terminal **104** are disposed in proximity, so that the total wiring length of the wiring line connecting the external output terminal **104** and the PA module **2B** and the wiring line connecting the external input terminal **105** and the PA module **2B** can be reduced. Therefore, the waveform distortion of the power supply voltage V.sub.ET output from the ET module **1C** can be suppressed, thereby suppressing the efficiency deterioration.

[0247] Furthermore, in the plan view of the module laminate **90**, the external output terminal **104** and the external input terminal **105** are adjacently disposed. This configuration further reduces the above total wiring length.

[0248] In addition, the external input terminal **107** is disposed adjacent to the side P3. As a result of this, the external input terminal **107** and the external output terminal **106** are disposed in proximity, so that the total wiring length of a wiring line connecting the external output terminal **106** and the PA module **2C** and a wiring line connecting the external input terminal **107** and the PA module **2C** can be reduced. Therefore, the waveform distortion of the power supply voltage V.sub.ET output from the ET module **1C** can be suppressed, thereby suppressing the efficiency deterioration.

[0249] Furthermore, in the plan view of the module laminate **90**, the external output terminal **106** and the external input terminal **107** are adjacently disposed. This configuration further reduces the above total wiring length.

[0250] In addition, the external output terminal **101** and the external input terminal **102** are connected to the capacitor **31A**. As a result of this configuration, when the APT mode is applied to the ET module **1C**, the capacitor **31A** can be configured to function as a bypass condenser, and the voltage level of the power supply voltage V.sub.ET can be stabilized.

[0251] In addition, the external output terminal **104** and the external input terminal **105** are connected to the capacitor **31B**. As a result of this configuration, when the APT mode is applied to the ET module **1C**, the capacitor **31B** can be configured to function as a bypass condenser, and the voltage level of the power supply voltage V.sub.ET can be stabilized.

[8. Technical Effects]

[0252] As described above, the ET module **1** according to the embodiment includes the module laminate **90**; the external output terminals **101** and **104** disposed on the module laminate **90**; and the voltage supply circuit **10** disposed on the module laminate **90** and configured to supply the power supply voltage V.sub.ETA based on an envelope signal to the PA module **2A** via the external output terminal **101**, and to simultaneously supply the power supply voltage V.sub.ETB based on an envelope signal to the PA module **2B** via the external output terminal **104**. The module laminate

90 has the side **P1** and the side **P2** adjacent to the side **P1**, the sides **P1** and **P2** forming at least a part of the outer shape of the module laminate **90** in plan view of the module laminate **90**. The external output terminal **101** is disposed adjacent to the side **P1**, and the external output terminal **104** is not disposed adjacent to the side **P1**, but is disposed adjacent to the side **P2**.

[0253] As a result of this, as compared to the configuration in which the external output terminals **101** and **104** are adjacently disposed on the same side of the module laminate **90**, the wiring lines supplying the power supply voltages $V_{sub.ETA}$ and $V_{sub.ETB}$ can be disposed at a distance from each other. Therefore, the heat dissipation of the ET module **1** can be improved, and the waveform distortion of the power supply voltages $V_{sub.ETA}$ and $V_{sub.ETB}$ caused by the heat generation can be suppressed, thereby suppressing the efficiency deterioration.

[0254] In addition, the ET module **1A** according to Example 1 includes the module laminate **90**; the external output terminals **101** and **104** disposed on the module laminate **90**; and the voltage supply circuit **10A** including at least one integrated circuit **80** disposed on the module laminate **90**. The voltage supply circuit **10A** includes at least one switch included in the switched-capacitor circuit **12**, at least one switch included in the output switching circuit **13A**, and at least one switch included in the output switching circuit **13B**. The switched-capacitor circuit **12** is configured to generate a plurality of discrete voltages based on an input voltage and to output the plurality of discrete voltages generated to the output switching circuits **13A** and **13B**. The output switching circuit **13A** is configured to selectively output at least one of the plurality of discrete voltages generated to the PA module **2A** via the external output terminal **101**, and the output switching circuit **13B** is configured to selectively output at least one of the plurality of discrete voltages generated to the PA module **2B** via the external output terminal **104**. The module laminate **90** has the side **P1** and the side **P2** adjacent to the side **P1**, the sides **P1** and **P2** forming at least a part of the outer shape of the module laminate **90** in plan view of the module laminate **90**. The external output terminal **101** is disposed adjacent to the side **P1**, and the external output terminal **104** is not disposed adjacent to the side **P1**, but is disposed adjacent to the side **P2**.

[0255] As a result of this, as compared to the configuration in which the external output terminals **101** and **104** are adjacently disposed on the same side of the module laminate **90**, the wiring lines supplying the power supply voltages $V_{sub.ETA}$ and $V_{sub.ETB}$ can be disposed at a distance from each other. Therefore, the heat dissipation of the ET module **1A** can be improved, and the waveform distortion of the power supply voltages $V_{sub.ETA}$ and $V_{sub.ETB}$ caused by the heat generation can be suppressed, thereby suppressing the efficiency deterioration.

[0256] In addition, for example, the ET modules **1**, **1A**, **1B**, and **1C** further include the external input terminal **102** connected to the first path connecting the external output terminal **101** and the PA module **2A**, and the external input terminal **102** is disposed adjacent to the side **P1**.

[0257] As a result of this, the external input terminal **102** and the external output terminal **101** are disposed in proximity, so that the total wiring length of the wiring line connecting the external output terminal **101** and the PA module **2A** and the wiring line connecting the external input terminal **102** and the PA module **2A** can be reduced. Therefore, the waveform distortion of the power supply voltage $V_{sub.ETA}$ output from the ET module can be suppressed, thereby suppressing the efficiency deterioration.

[0258] In addition, for example, in the ET modules **1**, **1A**, **1B**, and **1C**, the external output terminal **101** and the external input terminal **102** are adjacently disposed in the plan view of the module laminate **90**.

[0259] This configuration further reduces the above total wiring length.

[0260] In addition, for example, the ET modules **1**, **1A**, **1B**, and **1C** further include the external input terminal **105** connected to the second path connecting the external output terminal **104** and the PA module **2B**, and the external input terminal **105** is disposed adjacent to the side **P2**.

[0261] As a result of this, the external input terminal **105** and the external output terminal **104** are disposed in proximity, so that the total wiring length of the wiring line connecting the external

output terminal **104** and the PA module **2B** and the wiring line connecting the external input terminal **105** and the PA module **2B** can be reduced. Therefore, the waveform distortion of the power supply voltage $V_{sub.ETB}$ output from the ET module can be suppressed, thereby suppressing the efficiency deterioration.

[0262] Furthermore, for example, in the plan view of the module laminate **90**, the external output terminal **104** and the external input terminal **105** are adjacently disposed.

[0263] This configuration further reduces the above total wiring length.

[0264] In addition, for example, in the ET modules **1**, **1A**, **1B**, and **1C**, in the plan view of the module laminate **90**, the external connection terminal **150** (GND) is disposed adjacent to the external output terminal **101**, and the external connection terminal **150** (GND) is disposed adjacent to the external output terminal **104**.

[0265] As a result of this, interference between the power supply voltage $V_{sub.ETA}$ supplied via the external output terminal **101** and the power supply voltage $V_{sub.ETB}$ supplied via the external output terminal **104** can be suppressed, thereby suppressing the waveform distortion of the power supply voltage $V_{sub.ETA}$ and the power supply voltage $V_{sub.ETB}$.

[0266] In addition, for example, in the ET modules **1**, **1A**, **1B**, and **1C**, in the plan view of the module laminate **90**, the external connection terminal **150** (GND) may be disposed between the external output terminal **101** and the external output terminal **104**.

[0267] As a result of this, the interference between the power supply voltage $V_{sub.ETA}$ supplied via the external output terminal **101** and the power supply voltage $V_{sub.ETB}$ supplied via the external output terminal **104** can be suppressed, thereby suppressing the waveform distortion of the power supply voltage $V_{sub.ETA}$ and the power supply voltage $V_{sub.ETB}$.

[0268] In addition, for example, the ET modules **1A** and **1C** may further include a low pass filter connected between the output switching circuit **13A** and the external output terminal **101**.

[0269] This configuration removes noise due to pulse waveforms of the plurality of discrete voltages output from the output switching circuit **13A**.

[0270] In addition, for example, in the ET modules **1A** and **1C**, the external output terminal **101** and the external input terminal **102** are connected to the capacitor **31A**.

[0271] As a result of this configuration, when the APT mode is applied to the ET modules **1A** and **1C**, the capacitor **31A** can be configured to function as the bypass condenser, and the voltage level of the power supply voltage $V_{sub.ETA}$ can be stabilized.

[0272] In addition, for example, the ET modules **1A** and **1C** further includes the switch **21A** that switches between connection and disconnection between the external input terminal **102** and the ground.

[0273] As a result of this configuration, when the digital ET mode is applied to the ET modules **1A** and **1C**, the switch **21A** is in a non-conductive state, so that a radio frequency signal of the wide channel band width (wide modulation band width) can be amplified highly efficiently in the PA module **2A**. In addition, when the APT mode is applied to the ET modules **1A** and **1C**, the switch **21A** is in a conductive state, so that the capacitor **31A** can be configured to function as the bypass condenser, and the voltage level of the power supply voltage $V_{sub.ETA}$ can be stabilized.

[0274] In addition, for example, in the ET module **1**, at least one of the external input terminal **102** and the external output terminal **101** is connected to the inductor **33A**.

[0275] As a result of this configuration, when the analog ET mode is applied to the ET module **1i**, the inductor **33A** can be configured to function as the choke coil that blocks a radio frequency signal leaking from the PA module **2A**. Therefore, it is possible to return the feedback signal of the power supply voltage $V_{sub.ETA}$ on the first path to the ET module **1B** with high precision via the external input terminal **102**. In addition, the superimposition of the radio frequency noise on the power supply voltage $V_{sub.ETA}$ output from the external output terminal **101** can be suppressed.

[0276] In addition, for example, in the ET module **1B**, the external input terminal **102** is connected to the voltage supply circuit **10B**.

[0277] This makes it possible to return the feedback signal of the power supply voltage $V_{\text{sub.ETA}}$ on the first path to the voltage supply circuit **10B** via the external input terminal **102**.

[0278] In addition, the communication device **5** according to the embodiment includes the RFIC **4** that processes radio frequency signals, the ET module **1** connected to the RFIC **4**, the PA modules **2A** and **2B**, and the motherboard **92** on which the RFIC **4**, the ET module **1**, as well as the PA modules **2A** and **2B** are disposed.

[0279] This configuration realizes the effects of the ET module **1** in the communication device **5**.

[0280] In addition, for example, the communication device **5A** according to Example 1 (the communication device **5C** according to a modification example) further includes the capacitor **31A** that is disposed on the motherboard **92** and connected to the external output terminal **101**.

[0281] As a result of this configuration, when the APT mode is applied to the ET module **1A** and **1C**, the capacitor **31A** can be configured to function as the bypass condenser, and the voltage level of the power supply voltage $V_{\text{sub.ETA}}$ can be stabilized.

[0282] In addition, for example, the communication device **5B** according to Example 2 further includes the inductor **33A** that is disposed on the motherboard **92** and connected to the external output terminal **101**.

[0283] As a result of this configuration, when the analog ET mode is applied to the ET module **1i**, the inductor **33A** can be configured to function as the choke coil that blocks a radio frequency signal leaking from the PA module **2A**. Therefore, the superimposition of the radio frequency noise on the power supply voltage $V_{\text{sub.ETA}}$ output from the external output terminal **101** can be suppressed.

Additional Exemplary Embodiment

[0284] As described above, the tracker modules and the communication devices according to the present disclosure have been described based on the embodiments, examples, and modification examples. However, the tracker modules and the communication devices according to the exemplary aspects of the present disclosure are not limited to the above-described embodiments, examples, and modification examples. It should be appreciated that the present disclosure also includes other embodiments realized by combining any of the components in the above-described embodiments, examples, and modification examples, modification examples obtained by applying various modifications to the above-described embodiments, examples, and modification examples without departing from the spirit of the exemplary aspects of the present disclosure, or various types of devices that incorporate the tracker modules or the communication devices described above.

[0285] For example, in the circuit configurations of the various circuits according to the above-described embodiment, examples, and modification examples, another circuit element and wiring line, or the like may be inserted between the paths connecting the circuit elements and the signal paths disclosed in the drawings.

[0286] In the following, a description is given of the characteristics of the tracker modules and the communication devices described based on the embodiment described above.

[0287] <1> A tracker module is provided that includes a module laminate; a first external output terminal and a second external output terminal disposed on the module laminate; and a voltage supply circuit disposed on the module laminate and configured to supply a first voltage based on an envelope signal to a first power amplifier via the first external output terminal, and to simultaneously supply a second voltage based on an envelope signal to a second power amplifier via the second external output terminal. The module laminate has a first side and a second side adjacent to the first side. Moreover, the first side and the second side form at least a part of an outer shape of the module laminate in plan view of the module laminate. The first external output terminal is disposed adjacent to the first side, and the second external output terminal is disposed adjacent to the second side but not the first side.

[0288] <2> A tracker module is provided that includes a module laminate; a first external output

terminal and a second external output terminal disposed on the module laminate; and a voltage supply circuit including at least one integrated circuit disposed on the module laminate. The at least one integrated circuit includes at least one switch included in a switched-capacitor circuit, at least one switch included in a first output switching circuit, and at least one switch included in a second output switching circuit. The switched-capacitor circuit is configured to generate a plurality of discrete voltages based on an input voltage and to output the generated plurality of discrete voltages to the first output switching circuit and the second output switching circuit. The first output switching circuit is configured to selectively output at least one discrete voltage of the generated plurality of discrete voltages to a first power amplifier via the first external output terminal, and the second output switching circuit is configured to selectively output at least one discrete voltage of the generated plurality of discrete voltages to a second power amplifier via the second external output terminal. The module laminate has a first side and a second side adjacent to the first side. Moreover, the first side and the second side form at least a part of an outer shape of the module laminate in plan view of the module laminate. The first external output terminal is disposed adjacent to the first side, and the second external output terminal is disposed adjacent to the second side but not the first side.

[0289] <3> The tracker module according to <1> or <2>, further including a first external input terminal connected to a first path connecting the first external output terminal and the first power amplifier, in which the first external input terminal is disposed adjacent to the first side.

[0290] <4> The tracker module according to <3>, wherein, in the plan view of the module laminate, the first external output terminal and the first external input terminal are disposed adjacent to each other.

[0291] <5> The tracker module according to any one of <1> to <3>, further including a second external input terminal connected to a second path connecting the second external output terminal and the second power amplifier, in which the second external input terminal is disposed adjacent to the second side.

[0292] <6> The tracker module according to <5>, wherein, in the plan view of the module laminate, the second external output terminal and the second external input terminal are disposed adjacent to each other.

[0293] <7> The tracker module according to any one of <1> to <6>, wherein, in the plan view of the module laminate, a ground terminal is disposed adjacent to the first external output terminal, and a ground terminal is disposed adjacent to the second external output terminal.

[0294] <8> The tracker module according to any one of <1> to <6>, wherein, in the plan view of the module laminate, a ground terminal is disposed between the first external output terminal and the second external output terminal.

[0295] <9> The tracker module according to <2>, further including a low pass filter connected between the first output switching circuit and the first external output terminal.

[0296] <10> The tracker module according to <3>, in which the first external output terminal and the first external input terminal are connected to a capacitor.

[0297] <11> The tracker module according to <10>, further including a first switch circuit that switches between connection and disconnection between the first external input terminal and a ground.

[0298] <12> The tracker module according to <3>, in which at least one of the first external input terminal and the first external output terminal is connected to an inductor.

[0299] <13> The tracker module according to <3> or <12>, in which the first external input terminal is connected to the voltage supply circuit.

[0300] <14> A communication device is provided that includes a signal processing circuit that processes radio frequency signals; the tracker module according to any one of <1> to <13>, which is connected to the signal processing circuit; the first power amplifier and the second power amplifier; and a motherboard on which the signal processing circuit, the tracker module, and the

first power amplifier and the second power amplifier are disposed.

[0301] <15> The communication device according to <14>, further including a capacitor disposed on the motherboard and connected to the first external output terminal.

[0302] <16> The communication device according to <14>, further including an inductor disposed on the motherboard and connected to the first external output terminal.

[0303] The exemplary aspects of the present disclosure can be widely used in communication devices, such as mobile phones, as a tracker module that supplies voltages to a power amplifier.

REFERENCE SIGNS LIST

[0304] **1**, **1A**, **1B**, **1C** ET MODULE [0305] **2A**, **2B**, **2C** PA MODULE [0306] **3A**, **3B**, **3C** ANTENNA [0307] **4** RFIC [0308] **5**, **5A**, **5B**, **5C** COMMUNICATION DEVICE [0309] **10**, **10A**, **10B**, **10C** VOLTAGE SUPPLY CIRCUIT [0310] **11** PRE-REGULATOR CIRCUIT [0311] **11S** PR SWITCH PORTION [0312] **12** SWITCHED-CAPACITOR CIRCUIT [0313] **12S** SC SWITCH PORTION [0314] **13A**, **13B** OUTPUT SWITCHING CIRCUIT [0315] **13S** OS SWITCH PORTION [0316] **14A**, **14B** FILTER CIRCUIT [0317] **14S** FL SWITCH PORTION [0318] **15A**, **15B** DC-DC CONVERTER [0319] **16A**, **16B** LINEAR AMPLIFIER CIRCUIT [0320] **21A**, **21B**, **61**, **62** SWITCH [0321] **31A**, **31B** CAPACITOR [0322] **33A**, **33B** INDUCTOR [0323] **41**, **42**, **43**, **44**, **45**, **46** POWER AMPLIFIER [0324] **51**, **52**, **53**, **54**, **55**, **56** FILTER [0325] **70** CONTROL CIRCUIT [0326] **70A** DIGITAL CONTROL SECTION [0327] **80** INTEGRATED CIRCUITS [0328] **90** MODULE LAMINATE [0329] **90a**, **90b** MAIN SURFACE [0330] **91** RESIN MEMBER [0331] **92** MOTHERBOARD [0332] **101**, **104**, **106** EXTERNAL OUTPUT TERMINAL [0333] **102**, **105**, **107** EXTERNAL INPUT TERMINALS [0334] **103** CONTROL SIGNAL TERMINAL [0335] **110**, **131A**, **131B**, **132A**, **132B**, **133A**, **133B**, **134A**, **134B**, **140A**, **140B** INPUT TERMINALS [0336] **111**, **112**, **113**, **114**, **130A**, **130B** OUTPUT TERMINALS [0337] **150** EXTERNAL CONNECTION TERMINAL [0338] **P1**, **P2**, **P3**, **P4** SIDE

Claims

1. A tracker module comprising: a module laminate; a first external output terminal and a second external output terminal on the module laminate; and a voltage supply circuit on the module laminate and configured to supply a first voltage based on an envelope signal to a first power amplifier via the first external output terminal, and to simultaneously supply a second voltage based on an envelope signal to a second power amplifier via the second external output terminal, wherein the module laminate has a first side and a second side adjacent to the first side, wherein the first side and the second side form at least a part of an outer shape of the module laminate in a plan view of the module laminate, wherein the first external output terminal is disposed adjacent to the first side, and wherein the second external output terminal is disposed adjacent to the second side but is not disposed adjacent to the first side.
2. The tracker module according to claim 1, further comprising: a first external input terminal connected to a first path that connects the first external output terminal to the first power amplifier, wherein the first external input terminal is disposed adjacent to the first side.
3. The tracker module according to claim 2, wherein, in the plan view of the module laminate, the first external output terminal is adjacent to the first external input terminal.
4. The tracker module according to claim 1, further comprising: a second external input terminal connected to a second path that connects the second external output terminal to the second power amplifier, wherein the second external input terminal is disposed adjacent to the second side.
5. The tracker module according to claim 4, wherein, in the plan view of the module laminate, the second external output terminal is adjacent to the second external input terminal.
6. The tracker module according to claim 1, wherein, in the plan view of the module laminate, a ground terminal is disposed adjacent to the first external output terminal, and a ground terminal is disposed adjacent to the second external output terminal.

7. The tracker module according to claim 1, wherein in the plan view of the module laminate, a ground terminal is disposed between the first external output terminal and the second external output terminal.
8. The tracker module according to claim 2, wherein the first external output terminal and the first external input terminal are connected to a capacitor.
9. The tracker module according to claim 8, further comprising a first switch circuit configured to switch between a connection and a disconnection between the first external input terminal and a ground.
10. The tracker module according to claim 2, wherein at least one of the first external input terminal and the first external output terminal is connected to an inductor.
11. The tracker module according to claim 2, wherein the first external input terminal is connected to the voltage supply circuit.
12. A tracker module comprising: a module laminate; a first external output terminal and a second external output terminal disposed on the module laminate; and a voltage supply circuit including at least one integrated circuit on the module laminate, wherein the at least one integrated circuit includes at least one switch included in a switched-capacitor circuit, at least one switch included in a first output switching circuit, and at least one switch included in a second output switching circuit, wherein the switched-capacitor circuit is configured to generate a plurality of discrete voltages based on an input voltage and to output the generated plurality of discrete voltages to the first output switching circuit and the second output switching circuit, wherein the first output switching circuit is configured to selectively output at least one discrete voltage of the plurality of discrete voltages to a first power amplifier via the first external output terminal, wherein the second output switching circuit is configured to selectively output at least one discrete voltage of the plurality of discrete voltages to a second power amplifier via the second external output terminal, wherein the module laminate has a first side and a second side adjacent to the first side, wherein the first side and the second side form at least a part of an outer shape of the module laminate in a plan view of the module laminate, wherein the first external output terminal is disposed adjacent to the first side, and wherein the second external output terminal is disposed adjacent to the second side but is not disposed adjacent to the first side.
13. The tracker module according to claim 12, further comprising a low pass filter connected between the first output switching circuit and the first external output terminal.
14. The tracker module according to claim 12, further comprising: a first external input terminal connected to a first path that connects the first external output terminal to the first power amplifier, wherein the first external input terminal is disposed adjacent to the first side.
15. The tracker module according to claim 14, wherein, in the plan view of the module laminate, the first external output terminal is adjacent to the first external input terminal.
16. The tracker module according to claim 12, further comprising: a second external input terminal connected to a second path that connects the second external output terminal to the second power amplifier, wherein the second external input terminal is disposed adjacent to the second side.
17. The tracker module according to claim 16, wherein, in the plan view of the module laminate, the second external output terminal is adjacent to the second external input terminal.
18. A communication device comprising: a signal processing circuit configured to process radio frequency signals; the tracker module according to claim 1, the tracker module connected to the signal processing circuit; the first power amplifier and the second power amplifier; and a motherboard on which the signal processing circuit, the tracker module, and the first power amplifier and the second power amplifier are disposed.
19. The communication device according to claim 18, further comprising a capacitor disposed on the motherboard and connected to the first external output terminal.
20. The communication device according to claim 18, further comprising an inductor disposed on the motherboard and connected to the first external output terminal.

